

Multi-channel tests of the matter-bounce prediction $f_{\text{NL}}^{\text{local}} = -35/16$ and a joint $(r, f_{\text{NL}}^{\text{local}})$ no-go for single-field matter bounces at constant sound speed

Houston Golden^{1,*}

¹*Independent Researcher, Los Angeles, California, USA*

(Dated: September 22, 2026)

A dust-dominated contraction before a nonsingular bounce predicts a parameter-free local non-Gaussianity. A from-scratch, vertex-by-vertex in-in computation confirms $f_{\text{NL}}^{\text{local}} = -35/16 = -2.1875$ (Li *et al.* 2016), half Cai *et al.*'s (2009) value, whose slip is traced to amplitude normalization. A scheme-independent Bardeen-potential argument resolves the bounce continuation ambiguity in favor of scheme S2 on the Quintin-type background, under the raw-ADM cubic form giving $f_{\text{NL}}^{\text{after}} = -1.25$, and now extends to the LQC and poly backgrounds, which cross the same singular surface smoothly rather than by a jump: there the continuation is a one-parameter prescription (the Mukhanov–Sasaki operator is not essentially self-adjoint at the crossing), and its time-symmetric member gives a linear-transfer amplitude of -1.094 (LQC, with an independent prescription-free anchor) and -1.140 (poly, principal-value only), with the cubic term uncomputed there so the full $f_{\text{NL}}^{\text{after}}$ is only a bracket, $[-1.20, -0.09]$ and $[-1.27, -0.13]$ respectively (S1's superseded range on all three backgrounds was $[-0.65, -0.50]$). The tensor and S1 scalar transfers are identical ($r_{\text{after}}^{\text{S1}} = 16\epsilon = 24$, $\sim 670\times$ BICEP/Keck), but under the selected continuation the bounce amplifies r to $r_{\text{after}} \approx 9.4 \times 10^2$ (Quintin-type, $\sim 2.6 \times 10^4\times$ BICEP/Keck), 96.0 (LQC, $\sim 2.7 \times 10^3\times$), and 170.6 (poly, $\sim 4.7 \times 10^3\times$) – worse than S1's common $r_{\text{after}} = 24$ on every background. $n_s = 1$ exactly for pure dust (0.9649 anchors $w = -0.0029$); $n_T = -0.035$ is the one tilt falsifiable in principle. A reduced constant sound speed trades r for f_{NL} : $r = 24c_s$, while the transmitted and bounce terms grow as $1/c_s^2$ and $1/c_s^4$. Tensor viability needs $c_s < 1.5 \times 10^{-3}$, but tree level expires far above that ($1.2|f_{\text{NL}}|\zeta_{\text{rms}} \leq 1$ fails below $c_s \simeq 0.08$, where $r \simeq 1.8$ is already $49\times$ BICEP/Keck), and Planck's 95% interval needs $c_s \geq 0.525$ ($r \geq 12.6$). The windows are disjoint by $\sim 350\times$ in c_s : the single-field matter bounce, for the full $P(X)$ k -essence class at constant sound speed, is jointly excluded by r and f_{NL} here; r and the bounce term are exactly λ -independent (the cubic coefficient cancels by parity in the bounce window), the best case (DBI) giving $r_{\text{min}} = 10.3$ on the same 95% criterion. A time-dependent $c_s(\eta)$ is not covered. A curvaton-type spectator is (r, n_s) -viable but dilutes $-35/16$ by weight $(r/24)^2$, below a 1σ SPHEREx-class detection for every tensor-viable r ; a concrete prediction needs an unmodelled entropy sector. Three channels remain conditional checks on the excluded background: pulsar timing ($\gamma_{\text{pred}} = 5.07$ vs. NANOGrav's 3.2 ± 0.6 , a 90% half-width, i.e. a 5.1σ null), PBHs (ratio 1.84 ± 0.03 within this model's own γ_{cr} coverage, 6.7–7.0 decades short, $f_{\text{PBH}} = 0$), and SMBH seeds (FIRAS-excluded); DESI DR1 cannot separate candidates. The contribution is the exact amplitude, the λ -general no-go, and the surviving curvaton route.

I. INTRODUCTION

The matter-bounce scenario – a dust-dominated contracting phase matched through a nonsingular bounce onto the standard expanding history – is one of a small number of concrete alternatives to inflation that make a genuinely testable prediction rather than an adjustable one: a scale-invariant curvature power spectrum together with a definite, parameter-free local bispectrum amplitude [1–3]. Distinguishing bounce cosmology from inflation, or from Λ CDM with no primordial mechanism at all, is only possible where the two make different, checkable statements; a fixed number with no free parameter, measured in more than one independent channel, is among the strongest such statements available in the near term.

This paper does two things. Section II re-derives that number from scratch, independently of the two published

derivations that disagree by a factor of two, and locates the origin of the disagreement; Sec. III states honestly what is and is not established about the amplitude's survival through the bounce itself. Sections IV–VI then take the resulting value through three independent observational channels – pulsar timing, primordial black hole abundance, and the large-scale-structure bispectrum – each stated at exactly the strength its evidence supports. Every claim below is graded honestly: a reproduction is labeled reproduced, a first computation is labeled new, and a quoted number is labeled cited and never recomputed silently. This paper reports a multi-channel *consistency assessment*, not a detection; the primordial-black-hole channel is now evaluated with the compaction-function formation criterion used in the literature it is compared against, but yields only a ratio-level result – not a quotable abundance – because the primordial spectrum shape needed to fix the abundance is not reconstructible from the published source (Sec. V A).

* houston@hubify.com; ORCID: 0009-0008-5616-5994

II. THE EXACT MATTER-CONTRACTION AMPLITUDE

Cai *et al.* [1] print a squeezed-limit amplitude $f_{\text{NL}}^{\text{local}} = -35/8 = -4.375$ for the matter bounce. Li *et al.* [4], whose no-go analysis extends that of Quintin *et al.* [3], obtain exactly one half of it,

$$f_{\text{NL}}^{\text{local}} = -\frac{35}{16} = -2.1875. \quad (1)$$

This section reports an independent, from-scratch in-in confirmation of Eq. (1), locates the origin of the discrepancy with the printed $-35/8$, and states the open item this confirmation does not settle. The full computation, its two validation checks, and the code that produced them are archived and referenced in the reproducibility statement below.

A. Setup and validation

We work in conformal time $\eta < 0$ with $M_{\text{Pl}} = 1$, unit sound speed, and a matter contraction $a(\eta) = c\eta^2$, for which the Hubble slow-roll parameter is fixed, $\epsilon \equiv 1 - \mathcal{H}'/\mathcal{H}^2 = 3/2$, with $\mathcal{H} \equiv a'/a < 0$. The quadratic action for the comoving-gauge curvature perturbation ζ ($h_{ij} = a^2 e^{2\zeta} \delta_{ij}$, uniform-field gauge $\delta\phi = 0$) is $S_2 = \int d\eta d^3x a^2 \epsilon [\zeta'^2 - (\partial\zeta)^2]$, with Bunch–Davies mode functions fixed by the Wronskian normalization. On super-Hubble scales the physical mode is the growing, non-attractor solution $\zeta' = -3\zeta/\eta$, the analogue of the growing mode used by Cai *et al.* [1].

We define the local bispectrum amplitude in the isosceles squeezed configuration $k_1 \ll k_2 = k_3 \equiv k$ using the same definition as Refs. [1, 4] (the two differ in the evaluation of their shared amplitude normalization below, not in this definition),

$$f_{\text{NL}}^{\text{local}} \equiv \lim_{k_1 \rightarrow 0} \frac{10}{3} \frac{\mathcal{A}(k_1, k, k)}{2k^3}, \quad (2)$$

where $\mathcal{A}(k_1, k_2, k_3)$ is the in-in bispectrum amplitude in the normalization of Li *et al.* [4] Eq. (4.20) – the connected three-point function of ζ with the momentum-conserving delta function and the two-point normalization stripped, so that the general-configuration amplitude is $f_{\text{NL}} = (10/3)\mathcal{A}_{\text{tot}}/\sum_i k_i^3$ and the squeezed isosceles limit $\sum_i k_i^3 \rightarrow 2k^3$ gives the form above. Each vertex of Table I contributes additively to $\mathcal{A}_{\text{tot}}$ in this normalization; it is precisely this normalization step that Sec. II’s comparison identifies as the origin of the factor of two. (Sec. III reuses the symbol $\mathcal{A}$ for the matter-asymptotic scale-factor normalization; the two never appear in the same expression.) This is equivalent to the local template $\zeta = \zeta_g + \frac{3}{5}f_{\text{NL}}\zeta_g^2$, using the cubic action derived by Maldacena [5] in comoving gauge for a canonical field: two bulk pieces $\propto \epsilon^2$, one boundary/field-redefinition term, and two ϵ^3 pieces involving $\tilde{\chi} \equiv \partial^{-2}\zeta'$. All five are evaluated by first-order in-in perturbation theory, every Wick

TABLE I. Squeezed-limit contribution of each cubic vertex at $\epsilon = 3/2$, from a from-scratch in-in computation, to $f^{\text{sq}}(\mu) \equiv \lim_{k_1 \rightarrow 0} \frac{10}{3} \mathcal{A}(k_1, k, k; \mu)/(2k^3)$.

vertex	$f^{\text{sq}}(\mu)$
field redefinition (local + non-local part)	$-\frac{25}{16} - \frac{15}{16}\mu^2$
$\zeta\zeta'^2$ ($\epsilon^2 - \epsilon^3/2$)	$-\frac{32}{32}$
$\zeta(\partial\zeta)^2$ ^a	0
$\zeta'\partial\zeta\cdot\partial\tilde{\chi}$	$\frac{15}{8}\mu^2$
$\zeta(\partial_i\partial_j\tilde{\chi})^2$	$-\frac{15}{32}$
total	$-\frac{35}{16} + \frac{15}{16}\mu^2$

^a Zero at leading order $\mathcal{O}(k^2 S^2)$, where $S \equiv -\eta_*$ is the (positive) conformal time at which the vertex integral is evaluated, so that $k^2 S^2 = (k\eta_*)^2 \rightarrow 0$ in the squeezed, super-Hubble limit taken here.

contraction summed explicitly, every time integral performed exactly before taking $|k\eta_B| \rightarrow 0$. Before applying this pipeline to the matter contraction we reproduce two independent literature results with the same code: the de Sitter, constant- ϵ bispectrum of Ref. [5], matched term by term; and the ultra-slow-roll result of Namjoo, Firouzjahi and Sasaki [6], $f_{\text{NL}} = 5/2$. Both match exactly.

B. Result: a per-vertex table

Table I gives the squeezed-limit contribution of each vertex, from scratch, at $\epsilon = 3/2$, in the exact-isosceles configuration $k_2 = k_3$, for which $\mu \equiv \hat{\mathbf{k}}_1 \cdot \hat{\mathbf{k}} = 0$ identically. The total reproduces the $\mu = 0$ value

$$f_{\text{NL}}^{\text{local}}|_{k_1 \ll k_2 = k_3} = -\frac{35}{16} \quad (3)$$

– this is *not* the angular average $-15/8$ of Eq. (4) below. Relaxing $k_2 = k_3$ at $\mathcal{O}(k_1)$ and defining $\mu \equiv \hat{\mathbf{k}}_1 \cdot \hat{\mathbf{k}}$ as the cosine between the long mode and either (now near-equal) short leg gives, at general orientation,

$$f_{\text{NL}}^{\text{sq}}(\mu) = -\frac{35}{16} + \frac{15}{16}\mu^2, \quad (4)$$

a monopole $-15/8$ and a μ^2 coefficient $15/16$ (Legendre $\ell = 2$ coefficient $5/8$) under angular averaging. The equilateral and folded values are $-255/128$ and $-9/8$, matching Ref. [4]’s $c_s = 1$ limit exactly. Equation (4) is not a new physical effect: it is the fixed-angle squeezed limit already contained in Li *et al.*’s Eq. 4.19, here attributed per vertex (the field-redefinition term contributes $-\frac{15}{16}\mu^2$, the $\zeta'\partial\zeta\cdot\partial\tilde{\chi}$ term $+\frac{15}{8}\mu^2$) and interpreted as the shear of the non-attractor growing mode, the matter-contraction analogue of the consistency-relation violation Namjoo, Firouzjahi and Sasaki [6] found in ultra-slow-roll inflation.

C. The Cai et al. factor of two

Reading the printed shape function of Cai *et al.* [1] (their Eq. 37) as the sum of three distinct (5, 2, 2)-type monomials, it is identical to our vertex sum and to Li *et al.*'s Eq. 4.19 at $c_s = 1$ (difference zero, checked symbolically). Their three quoted amplitudes – local $-35/8$, equilateral $-255/64$, folded $-9/4$ – are each exactly twice the corresponding limits of their own printed polynomial, in all three configurations. We locate the discrepancy between their printed shape function and their quoted amplitudes at this normalization step; this localizes the factor of two to a point between their printed shape function and their quoted amplitudes, uniformly across their Eqs. (38)–(41) and their Fig. 5 (all four printed configurations), where the printed shape function is *effectively* converted to the quoted amplitude via $f_{\text{NL}}^{\text{local}} = (20/3)\mathcal{A}/\Sigma k^3$. Their printed shape function, Eq. (37), agrees with Li *et al.* [4]'s Eq. 4.19 *identically under the distinct-monomial reading above* (difference zero, checked symbolically); under the alternative six-permutation reading of the same sum the difference is $-(99/128)\Sigma k^3$ (squeezed limit $-305/64$), which is why the reading must be stated. Li *et al.* redo the in-in calculation for general sound speed and print the amplitude directly in their Eq. 5.1, which reduces to $-35/16$ at $c_s = 1$. Li *et al.* do not themselves comment on the discrepancy with Cai *et al.*'s quoted amplitude; because their shape function (Eq. 4.19) agrees with Cai *et al.*'s (Eq. 37) exactly, the localization above – a uniform factor-2 slip between Cai *et al.*'s printed shape function and their quoted amplitudes – is adjudicated directly from the two shape functions here, not a convention difference reported by either source. Which of the two steps carries the factor cannot be decided from print alone. Quintin *et al.* [3] likewise quote $-35/16$ in the text following their cubic-action equation for $\mathcal{L}_3$, explicitly attributing it to Cai *et al.* as a citation rather than an independent computation. Li *et al.* already give an independent in-in re-derivation of the amplitude itself (Eq. 5.1); the present computation's novelty relative to theirs is not the amplitude number but the per-vertex attribution of it (the per-vertex table above) and the factor-of-two localization above, neither of which either published derivation provides.

D. Cross-check by an independent classical route

A separate-universe (δN) calculation on the same background but evaluated on uniform-density slices gives, for general ϵ , $f_{\text{NL}}^\rho = 5(\epsilon - 7)/8$, which reduces at $\epsilon = 3/2$ to $(5\epsilon - 35)/8 = -55/16$, while the comoving-slice δN (an isotropic, shear-free approximation) gives $f_{\text{NL}}^c = -5$ for all ϵ . Both f_{NL}^ρ and f_{NL}^c are normalized against δN itself (the Maldacena convention, ζ_{Mald}), not against δN_c ; since $\delta N_c = (1 - \epsilon/3)\zeta_{\text{Mald}}$ (Eq. (5)) at $\epsilon = 3/2$ gives $1 - \epsilon/3 = 1/2$, an f_{NL} normalized against δN_c instead would differ from the values quoted here by a factor

$1/(1 - \epsilon/3) = 2$ at $\epsilon = 3/2$ – numerically the same factor of two as Cai *et al.*'s slip (Sec. II), but a different mechanism: this one is a normalization-convention choice between ζ_{Mald} and δN_c , not an amplitude-conversion error, and the coincidence is not pursued further here. Neither equals the in-in result above. An anisotropic (Bianchi-I) separate-universe treatment does not close the gap either: its projected shear response is traceless and supplies no monopole (Appendix A, Sec. A 4).

The resolution is that δN on the zero-shift threading computes a different variable from Maldacena's ζ . On a uniform- ϕ slice the comoving-gauge shift is $N_i = \partial_i \psi$ with $\psi \supset a^2 \epsilon \partial^{-2} \zeta$, which is $\mathcal{O}(1/k_L)$ in a non-attractor phase – the gradient-expansion assumption $N_i = \mathcal{O}(\nabla)$ that underlies every separate-universe construction fails here. Separating the long mode's coupling into its local response ([L]: via $\zeta_L, \dot{\zeta}_L$), its shear/curvature ([K]: via $\partial_i \partial_j \tilde{\chi}_L$, which cancels exactly between the varied legs of the quartic vertex), its *shift* ([X]: via $\partial_i \tilde{\chi}_L \propto 1/k_L$), and short-mode sourcing ([S], which is suppressed), the classical $\mathcal{O}(k^0)$ super-Hubble solution reproduces the in-in result of Eq. (4) per vertex and for the full shape function: [X] alone carries 100% of the quadrupole and $+5/4$ of the monopole, with the $1/k_L$ poles cancelling only in the sum over vertices – poles no separate-universe construction (isotropic or Bianchi-I) contains. The zero-shift-threading variable and ζ_{Mald} are related at linear order by

$$\delta N_c = \left(1 - \frac{\epsilon}{3}\right) \zeta_{\text{Mald}} \left(= \frac{1}{2} \zeta_{\text{Mald}} \text{ at } \epsilon = \frac{3}{2}\right), \quad (5)$$

derived from the Friedmann equation at fixed ϕ ($\delta H/H = -(\epsilon/3)\dot{\zeta}_L/H$) together with $\delta N_c = \int (K/3) d\tau$; at second order the map is a time-dependent, non-local (∂^{-2}) spatial diffeomorphism, so no local f_{NL} relation between δN_c and ζ_{Mald} exists, and comparing their squeezed limits directly (as the naive cross-check does) compares two different observables. The quadrupole gap is likewise closed by a computed identity, not merely bounded: the shift's dynamical coupling reproduces the anisotropic-rescaling ($\propto \mu^2$) piece exactly, and a final-time re-threading boundary term – absent from any separate-universe treatment – corrects it to the in-in coefficient exactly.

Scope statement.—The from-scratch in-in computation of Table I, independently fixing commutator, Wick-contraction, and orbit multiplicities from scratch and agreeing with two independently published results (Li *et al.*'s $c_s = 1$ limit and Quintin *et al.*'s quoted value), is the independent second-method route that adjudicates the factor of two *within the in-in method*, and it reproduces $-35/16$ exactly. The classical $\mathcal{O}(k^0)$ super-Hubble solution, organized by the shift decomposition above, provides a method-independent confirmation: it reproduces Eq. (4) exactly by a route independent of the in-in commutator expansion. The residual gap to every separate-universe δN value is explained at the equation level rather than reconciled away: at linear order the two calculations are related by the derived identity of

Eq. (5); at second order the zero-shift-threading variable δN_c and ζ_{Mald} are related by a time-dependent, non-local spatial diffeomorphism whose net second-order effect on the squeezed bispectrum Appendix A records as an identity, without deriving the mechanism that produces it – so the naive squeezed-limit comparison of the two calculations’ raw values compares two different variables, not two disagreeing predictions of the same one. This equation-level closure is established for the comoving-slice δN value (–5) and, by an independent derivation (the A2 lapse-monopole calculation, reproducibility statement below), for the uniform-density value as well: a direct second-order continuation of the uniform-density variable itself has squeezed monopole $f_{\text{NL}}^\rho = 5(\epsilon - 7)/8 = -55/16$ at dust, reproducing –55/16 quoted above exactly. The threading map’s long $\times$ short lapse monopole coefficient $A_2 = \epsilon(3 - \epsilon)^2/3$ is the correct constraint-solve value (not $2(3 - \epsilon)^2$); the earlier initial-label figure $5(2\epsilon - 15)/24 = -5/2$ came from composing the map with the linear-mode weight $\lambda' = 2\lambda$, whereas the ρ -surface time shift, acting on the second-order curvature perturbation, carries weight 3λ , and correcting the weight closes the 15/16 gap exactly. The in-in monopole –15/8 (dust, δN_c normalization), the comoving-slice δN value –5, and the uniform-density δN value –55/16 are therefore three well-defined variables related by exact threading maps, not competing claims about one number. The flagship in-in monopole –35/16 (Maldacena, ζ_{Mald} normalization) is unaffected. We adopt Eq. (1) throughout this paper as the value of ζ_{Mald} , the variable every downstream channel (transmission, PTA, PBH, LSS) inherits, and carry –35/8 in parallel wherever a channel distinguishes them. Full derivations, the general- ϵ formula, and the [L]/[K]/[X]/[S] decomposition are given in Appendix A and archived per the reproducibility statement below.

III. TRANSMISSION THROUGH THE BOUNCE

Equation (4) is a statement about the contraction phase alone, and it holds at every instant of that phase: Sec. II evaluates it in the limit $|k\eta_B| \rightarrow 0$ and its derivation is end-time independent at leading order (corrections are $\mathcal{O}(k^2\eta_B^2)$). Whether it survives into an observable post-bounce spectrum is a separate, narrower question that depends on physics not computed in full here, and we state the current evidence precisely under one explicit modeling assumption. (A4): the contraction-phase bispectrum is frozen at a handoff time η_h near the NEC boundary and evolved purely linearly thereafter – i.e. we bound the mode-mixing *transfer* of the already-fixed contraction-phase amplitude, not the amplitude itself, which Sec. II has already shown is independent of when in the contraction phase it is evaluated. For a linear mode-mixing matrix that carries the growing mode’s amplitude through the bounce with weight 1 and the sub-dominant decaying mode with

weight r_{mix} , writing $\rho(\eta_h) \equiv |J(-\eta_h)|/I_\infty \in (0, 1]$ for the fraction of the mode-mixing integral accumulated before η_h , the transmitted f_{NL} amplitude scales as $T_{f_{\text{NL}}}(\eta_h) = \frac{1 + r_{\text{mix}}[1 - \rho(\eta_h)]}{1 + 2r_{\text{mix}}}$, where $\mathcal{A}$ is the matter-asymptotic normalization of the scale factor ($a \rightarrow \mathcal{A}\tau^2$), $I_\infty \equiv J(+\infty)$ is the super-Hubble mode-mixing integral of the growing-mode basis, and $r_{\text{mix}} = -9i\mathcal{A}^2 I_\infty/k^3 = \beta_- I_\infty/\alpha_-$ is in general complex, with α_- , β_- the matter-basis projection coefficients and the denominator $1 + 2r_{\text{mix}}$ fixing the vacuum normalization $C_1 = \alpha_-(1 + r_{\text{mix}})$, $C_2 = \beta_-$; over the validated range $k\eta_B \lesssim 10^{-2}$ used throughout this paper, $|r_{\text{mix}}| \gg 1$ holds numerically for all three backgrounds below, and in this growing-mode-dominated limit the transfer reduces to

$$T_{f_{\text{NL}}} = \frac{1 - \rho(\eta_h)}{2}, \quad (6)$$

so that, under assumption (A4) and within scheme S1, $0 \leq T_{f_{\text{NL}}} < 1/2$ since $\rho \in (0, 1]$: within this handoff scheme, linear transfer can only suppress the amplitude, never invert or amplify its sign. This bound is a statement about scheme S1’s variable, which is exact only where $\epsilon = -\dot{H}/H^2$ is constant – true throughout the matter contraction but not through the bounce window itself – and on the Quintin-type background scheme S1 is replaced by the scheme-independent Bardeen-potential argument of Sec. III A, which selects S2; it should therefore not be read as this paper’s transfer bound on that background. S2’s raw-ADM continuation lies outside assumption (A4) and transmits with $T_{f_{\text{NL}}} \approx 1.03$ (from $|\lambda_\zeta| = 0.97$), outside $[0, 1/2]$. The general constant- c_s no-go of Sec. VIII retains the S1 assumption-(A4) framework, which the Bardeen argument has not yet been extended to cover at $c_s \neq 1$. $T_{f_{\text{NL}}}$ is precisely the f_{NL} transfer, $T_{f_{\text{NL}}} = 1/\lambda_\zeta$ with $\lambda_\zeta \equiv \zeta(+\infty)/\zeta(-\eta_h)$ the linear-amplitude growth of ζ across the handoff ($\lambda_\zeta = 4.0, 5.12, 6.06$ for the LQC, poly, and Quintin-type backgrounds respectively): ζ *grows* through the bounce by λ_ζ , which is why f_{NL} is suppressed rather than enhanced by a uniform rescaling. Separately, the post-bounce curvature power spectrum remains scale-invariant, $\Delta^2 \propto k^3 |\alpha_{\text{post}}|^2$ flat to 1.2–4.2% across the validated k range in the $|r_{\text{mix}}| \gg 1$ limit used here, so $n_s = 0.9649$ in Sec. VII is not in tension with this section. Evaluated across three explicit nonsingular bounce backgrounds – an LQC effective quasi-dust bounce (“LQC”), an analytic non-LQC polynomial bounce (“poly (analytic non-LQC)”, a different functional bounce profile from the LQC effective-dust form, not an LQC sub-case), and the Quintin *et al.* [3]-type parametrization (“Quintin-type”) – at the natural handoff $\eta_h = \eta_B$ (the NEC boundary), $T_{f_{\text{NL}}}$ ranges over 0.165–0.409 (the upper end being an S2, not an S1, linear-transfer value on LQC, not carried through cubic order and so not in Table III): a $1.64\times$ spread between two mode-function conventions (geometric vs. effective-fluid) on the *same* LQC background (0.409/0.250), and a $2.48\times$ spread across the full set of three backgrounds and

two conventions (0.409/0.165; the combined S1 range below uses the S1 rows only). The upper value, 0.409, is the LQC background under the effective-fluid convention (scheme S2); it is *not* a fourth background, and, as stated below, S2 has no computable *cubic-order* post-bounce $f_{\text{NL}}^{\text{after}}$ on the LQC background (the linear S2 transfer on LQC exists, $T = 0.409$; S2 is carried through cubic order only on the Quintin-type background, Sec. III A), so this LQC/S2 value does not enter the combined S1 range that follows (replaced on Quintin-type by the resolved S2 result below). This bound is a statement about scheme-dependent linear transfer, not a universal property of the physical post-bounce f_{NL} .

Direction of the $k\eta_B$ window.—The validity condition $k\eta_B \lesssim 10^{-2}$ used throughout this section is an *upper* bound on k , not a lower one: it is satisfied *most easily* at the largest observable scales (the CMB/LSS pivot, $k \sim 0.05 \text{ Mpc}^{-1}$) and *hardest* to satisfy at the smallest, PBH-relevant scales ($k \sim 10^{15}\text{--}10^{16} \text{ Mpc}^{-1}$ for a 10^{15} g PBH), where it requires a correspondingly high bounce energy (Sec. V C). Consequently the scheme-S1 transfer of this section is satisfied most comfortably at the LSS/CMB pivot used by Channel III below, and the transmitted amplitude $f_{\text{NL}}^{\text{after}} = T_{\text{fNL}} \cdot f_{\text{NL}}^{\text{before}} + \Delta f_{\text{NL}}^{\text{bounce}}$ (Sec. III A) is therefore this paper’s observable prediction for that channel, not the pre-bounce value.

What this bound does not include is behavior at $k\eta_B \gtrsim 10^{-2}$, outside the validated band, where three mechanisms have now been checked directly rather than left as a citation. **(1) Linear transfer at finite k .** Evaluated exactly (not in the super-Hubble limit) across $k\eta_B \in [0.1, 10]$ on all three backgrounds, the power-spectrum transfer relative to the S1 super-Hubble prediction is flat to 1–2% outside $k\eta_B \in [0.2, 2]$ but shows a real, background-dependent excursion, a factor 0.06–2.2 (not order-unity in the strict sense), peaking at $k\eta_B \approx 0.6\text{--}0.8$: the power ratio reaches 1.76 (Quintin-type), 2.24 (LQC), and 0.058 (poly non-LQC). **(2) The Quintin *et al.* scalar-field-velocity-dip amplification**, their Eq. (79): in their bounce-phase ansatz $H(t) = \Upsilon(t - t_B)$, $a(t) = a_B e^{\Upsilon(t - t_B)^2/2}$, $\Upsilon \equiv \dot{H}$ is the constant bounce-phase expansion rate and Δt_B the bounce duration in the NEC-violating window, related by $\Upsilon = 8/(3\Delta t_B^2)$; this paper’s grid point $\Delta t_B = 1$ gives $\Upsilon = 2.667$. Evaluated for this $(\Upsilon, \Delta t_B)$, the required amplification factor is exactly 1 on every background (constant on the Quintin-type background by construction, and not definable as a matter-sector effect on the two effective-fluid backgrounds), so no carrier for their mechanism exists here. **(3) The loop-quantum-cosmology cubic self-interaction enhancement** [7]: its third-order Hamiltonian is, by the authors’ own statement, the Legendre transform of the classical cubic action of Ref. [5] and contains no operator dependent on ρ/ρ_c or the dressed potential; every one of its nine terms maps exactly onto scheme S1’s vertex table; the operator is not absent from S1. Evaluated with the *exact* LQC-dust dressed-metric mode functions – not the super-Hubble reduction – across

$k\eta_B \in [0.1, 10]$ and three choices of initial vacuum state, the resulting bispectrum shows no counterpart of the literature’s order- 10^3 plateau and no feature at the literature’s own enhancement scale $k_{\text{LQC}}\eta_B \approx 1.06$: at and below k_{LQC} (squeezed configuration, $k\eta_B \lesssim 1.06$) it sits 3.1–4.4 decades below that plateau, with a state-dependence under 13%; this deficit is not extended above k_{LQC} , where the lab/ABS ratio changes sign (Table II). None of the three mechanisms reopens the pulsar-timing or PBH nulls (Secs. IV D, V C). Within scheme S1 and the (A4) handoff, the linear channel caps the transmission coefficient at 1/2, so a near-unity coefficient is not supported there; the scheme-S2 continuation selected in Sec. III A lies outside (A4) and transmits with a coefficient ≈ 1.03 on the Quintin-type background, so a near-unity coefficient *is* supported on that specific background under the selected scheme, but not within the S1/(A4) framework used for the LQC and poly backgrounds and for the $c_s \neq 1$ no-go of Sec. VIII.

TABLE II. Exact LQC-dust mode-function bispectrum $|f_{\text{NL}}^{\text{bounce}}|$ vs. the ABS literature plateau, lab-vacuum initial state (reproducibility statement below). The squeezed deficit is 3.1–4.4 dex only at and below $k_{\text{LQC}}\eta_B \approx 1.06$ (4.41/3.24/3.07 dex at $k\eta_B = 0.1/0.3/1$); above it the sign of the gap reverses.

configuration	$k\eta_B$	initial state	$ f_{\text{NL}}^{\text{bounce}} $
squeezed	0.1	lab vacuum	0.22
squeezed	0.3	lab vacuum	2.34
squeezed	1	lab vacuum	0.94
squeezed	3	lab vacuum	0.94
squeezed	10	lab vacuum	1.90
equilateral	0.1	lab vacuum	0.54
equilateral	0.3	lab vacuum	5.16
equilateral	1	lab vacuum	9.81
equilateral	3	lab vacuum	111.97
equilateral	10	lab vacuum	1215.57

A. The bounce’s own cubic term: now computed

The intrinsic bounce contribution left open above is closed for one explicit scheme. In the geometric (dressed-metric) prescription $z = a$, $\epsilon_{\text{eff}} = 1/2$, $c_s = 1$ – scheme S1, the substitution of these values into the classical comoving-gauge $P(X, \phi)$ cubic action of Ref. [5], an assumption of the scheme and not a derivation of the quantum-gravity-dressed operator – the squeezed-isosceles bounce contribution evaluated over the window $[-\eta_B, \eta_B]$ is, with $\rho_B \equiv \rho(\eta_B) = 1 - 2T_{\text{fNL}}$ the same handoff fraction of Sec. III evaluated at the bounce,

$$\Delta f_{\text{NL}}^{\text{bounce}}[\text{S1}] = -\frac{5}{24}\rho_B, \quad (7)$$

so that, combined with the linear transmission of the contraction-phase amplitude below, $f_{\text{NL}}^{\text{after}} = -\frac{85}{48}T_{f_{\text{NL}}} - \frac{5}{24}$, a closed-form vertex-table result confirmed to 3×10^{-4} by an independent finite- k numerical in-in evaluation of all six cubic attachments and boundary terms (Table I's five rows with the field-redefinition row's local and non-local parts counted separately, as the numerical pipeline treats them); the vertex $\zeta\dot{\zeta}^2$ carries 99.95–99.97% of it. Combining Eq. (7) with the linear transmission of Eq. (1) at $\eta_h = \eta_B$, $f_{\text{NL}}^{\text{after}} = T_{f_{\text{NL}}} \cdot (-35/16) + \Delta f_{\text{NL}}^{\text{bounce}}$, gives a definite post-bounce prediction for the first time, one number per background: The LQC/S2

TABLE III. Per-background transmitted amplitude. The first three rows are scheme S1 ($z = a$, Sec. III), superseded on every background by a Bardeen-potential continuation (Sec. III A) and retained here for comparison only. The fourth row, scheme S2 (Quintin-type-only raw-ADM continuation), is this paper's selected result on that background; its middle column is the net bounce-plus-expansion cubic contribution actually obtained (not a separately regulated $\Delta f_{\text{NL}}^{\text{bounce}}[\text{S2}]$ analogous to the S1 rows – see Sec. III A), so $f_{\text{NL}}^{\text{after}}[\text{S2}]$ is the full raw-ADM in-in result rather than $T_{f_{\text{NL}}} \cdot f_{\text{NL}}^{\text{before}} + \Delta$. The fifth and sixth rows extend the same Bardeen continuation to the LQC and poly backgrounds under the time-symmetric (principal-value) self-adjoint extension; their $\Delta f_{\text{NL}}^{\text{bounce}}$ is not computed, so $f_{\text{NL}}^{\text{after}}$ is reported only as a bracket between the two cubic endpoints this program has computed anywhere, never as a single value. On LQC the continuation carries a second, prescription-free anchor (the matter-kinetic weight has no zero there) and $T_{f_{\text{NL}}} = 1/2$ exactly, independent of prescription; on poly no such anchor exists and $T_{f_{\text{NL}}} = 0.521$ is the principal-value member of a one-parameter self-adjoint-extension family (a member one unit away in the extension parameter returns it to the S1 value, erasing the effect entirely). Numerical accuracy: the closed form for $\Delta f_{\text{NL}}^{\text{bounce}}$ reproduces the full in-in integral to 3×10^{-4} and its $c_s = 1$ gate to 4×10^{-6} on all three backgrounds, and the $T_{f_{\text{NL}}}$ integrations are step-converged to 1.5×10^{-8} or better (reproducibility statement below), so every digit quoted here is stable.

background	$T_{f_{\text{NL}}}$	$\Delta f_{\text{NL}}^{\text{bounce}}$	$f_{\text{NL}}^{\text{after}}$
Quintin-type [3] (S1)	0.165	−0.140	−0.501
LQC effective dust (S1)	0.250	−0.104	−0.651
poly (analytic non-LQC) (S1)	0.196	−0.127	−0.555
Quintin-type (Bardeen/S2)	1.03	+1.00	−1.25
LQC effective dust (Bardeen)	0.500	–	[−1.20, −0.09]
poly (Bardeen)	0.521	–	[−1.27, −0.13]

(effective-fluid) value $T_{f_{\text{NL}}} = 0.409$ has no row above: S2 has been carried through cubic order only on the Quintin-type background (below), so $|f_{\text{NL}}^{\text{after}}| \in [0.50, 0.65]$ (S1, all three backgrounds; the Quintin-type entry replaced by the S2 result below) against $|f_{\text{NL}}^{\text{before}}| = 2.19$: the bounce's own term is negative – it enhances $|f_{\text{NL}}|$, partially undoing but not reversing the linear suppression, at 19–39% of the transmitted contraction term. This range is valid only for $k\eta_B \lesssim 10^{-2}$ (set by the competing requirements that the post-bounce integral converge and the

super-Hubble boundary treatment hold) and only within scheme S1. This S1 range is superseded on every background by the Bardeen continuation below: exactly, by the single value $f_{\text{NL}}^{\text{after}} = -1.25$, on Quintin-type; for the linear transfer only, by -1.094 (LQC) and -1.140 (poly), on the other two, where the cubic term remains uncomputed (Table III).

Scheme S2.—The effective-fluid scheme S2's cubic bounce-window integral is finite: its apparent divergence is an artefact of the standard integrated-by-parts (Maldacena/Chen) form of the cubic action, which introduces explicit $1/H$ factors singular exactly at the bounce; on the exact linear S2 mode functions the comoving-gauge lapse and shift, $N_1 = \dot{\zeta}/H$ and $\psi = -\zeta/H + \chi$, are individually singular but their sum is regular at $H = 0$ (a residue cancellation), so the raw ADM cubic Lagrangian, taken before any integration by parts, is finite on-shell through the bounce and requires no cutoff. Evaluated on this raw form with the exact S2 modes on the Quintin-type background, the bounce-window in-in integral gives, at $k\eta_B = 10^{-3}, 3 \times 10^{-3}, 10^{-2}$ respectively, $f_{\text{NL}}^{\text{after}}[\text{S2}] = -1.249, -1.249, -1.246$ against $f_{\text{NL}}^{\text{before}} = -2.187 \approx -35/16$ – a factor ≈ 2.5 above S1's -0.501 on the same background. The difference is dominated by the choice of linear Mukhanov–Sasaki variable, not by any single cubic operator: S2 transmits the growing mode almost unsuppressed ($|\lambda_\zeta| = 0.97$) where S1 amplifies it by 6.06, and the net bounce-plus-expansion cubic contribution is $+1.0$ (S2) against -0.14 (S1), traced by an operator-by-operator decomposition to a geometric-versus-lapse cancellation rather than a single dominant vertex. S1 and S2 are not two regularizations of one answer, but neither are they two equally admissible continuations: the ambiguity is resolvable. The Bardeen potential Φ obeys $\Phi'' + 2(\mathcal{H} - \phi''/\phi')\Phi' + (k^2 + 2\mathcal{H}' - 2\mathcal{H}\phi''/\phi')\Phi = 0$, which contains neither a choice of z nor any explicit $1/H$; the friction coefficient $2(\mathcal{H} - \phi''/\phi')$ vanishes identically at $H = 0$ (verified symbolically), reducing the equation there to $\Phi'' + (k^2 + 2a^2\Upsilon)\Phi = 0$, manifestly regular; at a simple zero of $\dot{H}$ (the NEC crossing $\rho + p = 0$, where z_{S2}^2 vanishes and ζ is logarithmically divergent) the equation's indicial exponents are $\{0, 2\}$, both non-negative, so Φ admits a regular (Frobenius) solution branch there as well, for any smooth bounce – this is what “equally regular” means: a regular singular point with a non-singular solution, not an absence of singular coefficients. Written as the first-order system $\Phi' = -\dot{H}\Xi - \mathcal{H}\Phi$, $\Xi' = (a^2 - \varpi k^2)\Phi + \mathcal{H}\Xi$ with $\Xi \equiv \varpi(\Phi' + \mathcal{H}\Phi)$ and $\varpi = a^2/(\mathcal{H}^2 - \mathcal{H}') = -1/\dot{H}$ (not the squeezed-angle μ of Sec. II), it is free of δ -functions at $H = 0$, so Φ and Ξ are continuous across that boundary and no junction prescription need be imposed there – the continuity it enforces is identically S2's $[\zeta] = 0$, $[z^2\zeta'] = 0$. The Quintin-type background used below reaches this conclusion by jumping $\dot{H}$ discontinuously rather than by crossing $\rho + p = 0$ smoothly; replacing the jump with a C^2 blend of half-width d makes the background genuinely cross $\rho + p = 0$, and Richardson-

extrapolating the principal-value integral to $d \rightarrow 0$ reproduces the sharp-background $|\lambda_\zeta|$ below to 6×10^{-5} , 7×10^{-6} , and 5×10^{-6} at the three $k\eta_B$ values – a quantitative, not merely asserted, confirmation that the smooth limit agrees with the sharp result. The LQC and poly backgrounds are specified as effective fluids with no explicit scalar field ϕ and reach $\dot{H} = 0$ as an intrinsic feature of their own functional form rather than by a jump; the indicial argument above applies to them identically, and unlike the Quintin-type background – where $\dot{H}$ jumps and no $\rho + p = 0$ crossing exists – both cross $\rho + p = 0$ *smoothly*: at the crossing Φ and Φ' stay continuous, every solution obeying the degenerate momentum constraint $\Phi'(\eta_c) = -\mathcal{H}(\eta_c)\Phi(\eta_c)$ (the equation's own first recursion, not assumed), while the resonance between the indicial exponents $\{0, 2\}$ gives ζ a logarithmic divergence there with closed-form amplitude $c_{\log} = -k^2 \mathcal{H}(\eta_c) \Phi(\eta_c) / Q'(\eta_c)$, $Q \equiv \mathcal{H}^2 - \mathcal{H}' = -a^2 \dot{H}$; a logarithm is integrable, so a continuation through the crossing exists, and we take its symmetric (principal-value) finite part. Comparing at fixed incoming adiabatic vacuum via a handoff in the dust contraction, where the Bardeen and S1 ζ are the same function to 7.7×10^{-10} (LQC) and 1.3×10^{-8} (poly), this continuation transmits a fraction $R = 1/2$ (LQC) and $R = 3/8$ (poly) of scheme S1's amplitude, against $R = 0.160$ on the Quintin-type background; both rationals are exact, obtained independently by a finite- k two-sided principal-value regulator, a principal-value quadrature, and the closed form $R = 3I_\epsilon / I_{S1}$, $I_s \equiv \int d\eta / z_s^2$. Because $f_{\text{NL}} \propto 1/\zeta$ under a common linear rescaling, the transmitted amplitude scales as $1/R$, so scheme S1 – which transmits more curvature perturbation than the regular continuation on every background tested – *understates* $|f_{\text{NL}}^{\text{after}}|$: by $6.25 \times$ on the Quintin-type background, but only $2.00 \times$ (LQC) and $2.67 \times$ (poly); the factor of 6.25 quoted above is specific to the Quintin-type background and must not be carried to the other two. Unlike the Quintin-type handoff, which sits at the end of the exact dust contraction where ζ is unambiguous, the LQC/poly crossing lies inside the bounce region and is exactly where ζ diverges in this continuation; R is used precisely because it requires no handoff surface at all.

This continuation is a *prescription*, not a unique result, and the freedom is $O(1)$. At a simple sign-changing zero of $z^2 = 2a^2\epsilon$, writing $u \equiv \eta - \eta_c$, the Mukhanov–Sasaki operator has $z''/z = -1/(4u^2)$ – the critical inverse-square potential – and is not essentially self-adjoint: it admits a one-parameter family of self-adjoint extensions per surface, and continuity of Φ, Φ' does not select one. A complex-contour member of that family gives $R = 0.707$ (LQC) and $R = 0.750$ (poly), not the principal value, and a real extension parameter $\theta = \pi$ (LQC) or $\theta = 3.02$ (poly) returns R to 1, erasing the difference from scheme S1 entirely. On LQC this ambiguity is resolved independently of the extension parameter: the crossing is where the quantum-geometry factor $(1 - 2x)$ vanishes, not where the matter kinetic term does (the dust source

has $\rho + p = x \neq 0$ there), so the matter-built weight $z_K^2 = a^2(\rho + p)/(c_s^2 H^2)$ has no zero at all, its continuation is unambiguous, and its mixing integral coincides exactly with the principal value – giving $R = 1/2$ with no prescription. This anchor assumes the effective theory producing $H^2 = (\rho/3)(1 - \rho/\rho_c)$ leaves the scalar kinetic weight proportional to $a^2(\rho + p)/H^2$; were its constraint algebra instead to rescale that weight by the same $(1 - 2x)$ factor – plausible, since in holonomy-corrected effective dynamics that factor also multiplies k^2 and is associated with a signature change rather than ordinary evolution – the anchor would disappear and LQC would be as prescription-dependent as poly, for which no such anchor exists and $R = 3/8$ is quoted as the principal-value member only. This non-uniqueness was identified by an independent cross-check, carried out with no access to this derivation or its conclusion, which confirmed both rationals by an unrelated route (the same $\rho + p = x$ vs. $-2\dot{H} = x(1 - 2x)$ structural fact and the same partial-fraction decomposition of the poly integrand); an earlier version of this argument had wrongly treated the continuation as unique (reproducibility statement below).

The linear transfer alone moves $f_{\text{NL}}^{\text{after}, \text{lin}}$ from -0.547 to -1.094 (LQC) and from -0.428 to -1.140 (poly); the cubic-order term $\Delta f_{\text{NL}}^{\text{bounce}}$ on these two backgrounds is not computed by this program – it needs the raw-ADM in-in integral there, whose integrand carries the same $Q = 0$ logarithm at cubic order – so the full $f_{\text{NL}}^{\text{after}}$ on LQC and poly is reported in Table III only as a bracket between the two cubic endpoints this program has actually computed anywhere, never as a single value. With the tensor transfer scheme-independent (Sec. VII), $r_{\text{after}} = 24/R^2$ gives $r_{\text{after}} = 96.0$ (LQC) and 170.6 (poly) against the common scheme-S1 value 24.0 – exceeding the BICEP/Keck bound by $2.7 \times 10^3 \times$ and $4.7 \times 10^3 \times$ respectively, rather than by $6.7 \times 10^2 \times$: the tensor no-go of Sec. VIII is unchanged in direction and strengthened on both backgrounds (Sec. VII). Propagating this system across the Quintin-type bounce gives $|\lambda_\zeta| = 0.9699$ at $k\eta_B = 10^{-3}$ – 10^{-2} , agreeing with scheme S2 to 7×10^{-8} and smaller than S1's 6.06 by a factor 6.25. Scheme S1's $z = a$ is exact if and only if $\epsilon = -\dot{H}/H^2$ is constant – true throughout the matter contraction, where the two schemes coincide and both give $f_{\text{NL}}^{\text{before}} = -35/16$, and false in the bounce window, which is precisely where they part company. The transmitted amplitude on this background is therefore the single value $f_{\text{NL}}^{\text{after}} = -1.25$, not a band (the S1 range across all three backgrounds, $[-0.65, -0.50]$, is superseded on every background by the Bardeen continuation above – exactly, by -1.25 , on Quintin-type, and for the linear transfer only, by -1.094 and -1.140 , on LQC and poly, where the cubic term remains uncomputed and only a bracket is reported, Table III). A second, independent derivation, carried out without access to this argument or its conclusion, confirms S2, $\lambda_\zeta = 0.970$, by a different construction (the (Ψ, D) system with junctions fixed by Israel matching), reproducing the exact

ζ' identity, the constant- ϵ exactness condition, and the logarithmic divergence of ζ at a smooth NEC crossing (reproducibility statement below). Because the raw-form S2 bounce-window integral is not bounce-localised (an $O(1)$ dependence on the evaluation-window convention, unlike S1's), only $f_{\text{NL}}^{\text{before}}$ and $f_{\text{NL}}^{\text{after}}$ are convention-free S2 quantities; no separately regulated $\Delta f_{\text{NL}}^{\text{bounce}}[\text{S2}]$ analogous to Eq. (7) is reported, and S2's row for the Cai amplitude $-35/8$ is not computed (Table VI). The quoted $f_{\text{NL}}^{\text{after}}[\text{S2}]$ values use a uniform final evaluation time $\eta_*/\eta_B = 15$ at every k -point, which a 12-point scan (reproducibility statement below, `eta_star_scan`) shows lies in the stationary region of $f_{\text{NL}}^{\text{after}}$ at each of the three k -points and satisfies $k\eta_* \leq 0.15$ throughout. With this convention $f_{\text{NL}}^{\text{after}}[\text{S2}] = -1.2492, -1.2490, -1.2464$ at $k\eta_B = 10^{-3}, 3 \times 10^{-3}, 10^{-2}$, a spread of 0.22% across k and across the stationary window, so the headline -1.25 survives to three significant figures with a stated 0.3% evaluation-time systematic (a fourth digit is not justified).

Quintin *et al.*'s reported bispectrum growth through a nonsingular bounce [3] is not contradicted: their growth is the *additive* bounce-generated term, the counterpart of Eq. (7), sharing its sign (both enhance $|f_{\text{NL}}|$); the factor-of-two suppression identified above is the *multiplicative* transmission of the inherited contraction amplitude, a quantity they do not compute, so the two statements are about different terms and are not in tension. The nonlinear channel therefore has a single selected value on the Quintin-type background, $f_{\text{NL}}^{\text{after}} = -1.25$; the scheme qualifier is no longer needed for the linear transfer there, but is still owed for the cubic-action form (raw-ADM vs. Maldacena-form, above) and for the LQC and poly backgrounds, where the linear transfer is now resolved by the same argument but the cubic term is not, so only a bracket is reported (Table III). The absence of an orders-of-magnitude enhancement holds in both schemes (Sec. III). The full vertex table, regularization-assumption note, numerical cross-check, and literature-comparison derivation are archived and referenced in the reproducibility statement below.

IV. CHANNEL I: THE PULSAR-TIMING SPECTRAL SLOPE

A. Spectral conventions

With the characteristic strain parameterised as $h_c(f) \propto f^{(3-\gamma)/2}$ and $\Omega_{\text{GW}} = (2\pi^2/3H_0^2) f^2 h_c^2(f)$, the energy density scales as

$$\Omega_{\text{GW}}(f) \propto f^{5-\gamma}. \quad (8)$$

Hence $\gamma = 13/3$ for gravitational-wave-driven supermassive-black-hole binary inspirals ($\Omega_{\text{GW}} \propto f^{2/3}$), $\gamma = 5$ for a scale-invariant primordial tensor background ($n_T = 0$, $\Omega_{\text{GW}} \propto f^0$), and $\gamma = 2$ for $\Omega_{\text{GW}} \propto f^3$, the universal causality-limited infrared tail of an induced

background sourced by a finite-duration process [9]. $\gamma = 3$ ($\Omega_{\text{GW}} \propto f^2$) is the slope Papanikolaou [8] derives for a matter bounce whose source spectrum carries a specific near-linear low- k growth, $P_{\mathcal{R}} \propto k$; we retain it below only as a contrast case, since §IV D shows this model's own spectrum does not have that growth and does not reproduce $\gamma = 3$.

B. Official posterior and this paper's refit

The primary comparison for this channel is NANOGrav's own published result: fitting the 15-yr Hellings-Downs-correlated free-spectrum posterior on the lowest 14 frequency bins with a power law, NANOGrav report $\gamma_{\text{HD}} = 3.2_{-0.6}^{+0.6}$, quoted explicitly as a posterior median and 5–95% interval [11]; a Gaussian-equivalent $1\sigma \approx 0.6/1.645 \approx 0.365$. Against this official posterior, $\gamma = 3$ (matter-bounce, $\Omega_{\text{GW}} \propto f^2$) sits at 0.55σ , $\gamma = 5$ (scale-invariant primordial tensors, $n_T = 0$) at 4.94σ , and $\gamma = 13/3$ (SMBHB) at 3.1σ – consistent with NANOGrav's own characterization of 13/3 as in "moderate tension ... at the 99% credible boundary."

As a secondary, differently-conditioned cross-check, we independently refit the same Ceffyl kernel-density free-spectrum posteriors using *all* 30 frequency bins [11, 12] ($T_{\text{obs}} = 16.03$ yr) with a power law ($\gamma, \log_{10} A$) under priors $\gamma \sim U[0, 7]$, $\log_{10} A \sim U[-18, -11]$, using 32 walkers and 10^4 production steps after 2500 burn-in (3.2×10^5 samples; integrated autocorrelation time $\tau = (58.1, 58.0)$ for $(\gamma, \log_{10} A)$, giving $\text{ESS} = N/\max(\tau) = 5.5 \times 10^3$). The marginal is

$$\gamma = 2.567 \pm 0.382, \quad \log_{10} A = -14.03 \pm 0.38, \quad (9)$$

with median 2.591 and 68% interval [2.304, 2.882]. The marginal is left-skewed (skewness -1.10 , chain minimum 0.150), so its standard deviation exceeds the 68% half-width 0.289; the quoted ± 0.382 is the standard deviation throughout. This refit pipeline is validated by injection (Sec. IV C) and sits 1.20σ from the official posterior (quadrature-combined $\sigma = 0.53$ from the two marginals' widths). Table IV reports both the official-posterior tensions and Gaussian z -distances against the refit's own marginal, plus Savage-Dickey density ratios against the refit, $B_{\gamma_*/\text{free}} = p(\gamma_*/d)/\pi(\gamma_*)$ with $\pi = 1/7$, quoted only where the chain has posterior support: with zero of 3.2×10^5 samples at $\gamma \geq 5$ (chain maximum 4.70), no Bayes factor at $\gamma = 5$ is reported – a KDE evaluated there is extrapolating into an unsampled tail, and its value moves by an order of magnitude under a different bandwidth choice. The $\gamma = 13/3$ factor rests on 9 samples at $\gamma \geq 13/3$ and is quoted to one significant figure for the same reason.

The 1.14σ and 4.63σ refit entries are Gaussian approximations to a marginal that is not itself Gaussian (it is bounded by the sampling prior); directly from the chain, $P(\gamma > 3) = 8.97\%$. Narrowing the refit's γ prior from $U[0, 7]$ to $U[2, 5]$ moves $B_{3/\text{free}}$ from 3.23 to 1.47

TABLE IV. Channel I. Tensions against the official NANOGrav 14-bin posterior ($\gamma_{\text{HD}} = 3.2^{+0.6}_{-0.6}$, $\sigma \approx 0.365$) and against this paper’s 30-bin refit (Eq. (9)), with the refit’s Savage–Dickey factor where the chain has posterior support; the official posterior is the primary comparison and the refit is a secondary, differently-conditioned cross-check, not directly comparable to it. The causal floor ($\gamma = 2$, $\Omega_{\text{GW}} \propto f^3$, the universal IR bound discussed below in this section) is included as a comparator alongside the three physical hypotheses. All values reproduced from the archived chain – the author’s own 2026-05-01 reduction of the NANOGrav 15-yr KDE free spectra, so “reproduced” means re-read from that chain, not re-sampled (reproducibility statement below); differences from a self-reproduction run of the same script against the same chain are $\leq 3 \times 10^{-15}$ (floating-point-level agreement, not an independent external check). z_{refit} at $\gamma = 5$ is a Gaussian approximation only: 0 of 3.2×10^5 chain samples reach $\gamma \geq 5$ (chain max 4.70), so no data-supported Bayes factor is reported there (see text). The “Prim. tensors” row’s $\gamma = 5$ is the scale-invariant reference value; this model’s own first-order tensor background instead predicts $\gamma_{\text{pred}}^{(1)} = 5.035$ (reproducibility statement below), within 0.04 of it and subject to the same null verdict below.

Hyp.	γ_*	slope	z_{off}	z_{refit}	$B_{\gamma_*/\text{free}}$
Causal floor	2	f^3	3.29σ	1.48σ	0.54
Matter bounce, borrowed $\gamma = 3$	3	f^2	0.55σ	1.14σ	3.23
SMBHB	13/3	$f^{2/3}$	3.1σ	4.63σ	5×10^{-4}
Prim. tensors	5	f^0	4.9σ	6.37σ	n/a

while leaving $B_{\text{MB/SMBHB}} = B_{3/\text{free}}/B_{13/3/\text{free}} \approx 7 \times 10^3$ ($\log_{10} B \approx +3.9 \pm 0.2$, one significant figure given the 9-sample tail estimate) stable in the range $7\text{--}9 \times 10^3$; the individual Savage–Dickey factors $B_{\gamma_*/\text{free}}$ against the refit’s own prior (“nested” since each γ_* is a point restriction within the refit’s free- γ model) are prior-sensitive, but their ratio $B_{\text{MB/SMBHB}}$ (the model ratio) is not.

C. Refit validation

The refit’s likelihood/prior construction (30 bins, $T_{\text{obs}} = 16.03\text{ yr}$, $\gamma \sim U[0, 7]$, $\log_{10} A \sim U[-18, -11]$) is validated by an injection-recovery test that reuses that construction verbatim, using the REAL NANOGrav 15-yr HD-correlated free-spectrum KDE density grids from the public Zenodo record [10.5281/zenodo.8060824](https://zenodo.org/record/10.5281/zenodo.8060824) (30f_fs{hd}_ceffyl/*.npz, sha256-verified). Each bin’s real observed KDE curve (shape/width/skew preserved exactly) is re-centered along the $\log_{10} \rho$ axis to the model prediction at a chosen injected ($\gamma_{\text{true}}, \log_{10} A_{\text{true}}$), since a real dataset’s true signal is unknown and a ground-truth injection test requires a known target; this is the strongest injection achievable from real data without simulating a full new PTA dataset. At $\gamma_{\text{true}} = 13/3$, five realizations (the full 30-bin recovery plus four bootstrap bin-resamples) recover $\gamma = 4.336$ on average with a mean pull of $+0.016 \pm 0.044\sigma$ (std 0.0993 over the 5 realizations, standard error of the mean); at a $\gamma_{\text{true}} = 3$

control, five realizations recover $\gamma = 3.005$ with a mean pull of $+0.033 \pm 0.054\sigma$ (std 0.1203; script, JSON, and data manifest recorded in the reproducibility statement below). Both are consistent with unbiased recovery: both mean pulls are well under 0.1σ , though the per-realization scatter (std $\approx 0.10\text{--}0.12\sigma$ over 5 realizations) is comparable to, not smaller than, that threshold, tightening a secondary synthetic-density cross-check that substitutes a synthetic per-bin Gaussian density for the real KDE grids: $\gamma_{\text{true}} = 13/3 \rightarrow$ mean pull -0.026σ ; $\gamma_{\text{true}} = 3 \rightarrow$ mean pull $+0.068\sigma$ – both runs agree the pipeline is unbiased at well under 0.1σ . What this test establishes is *bias*, not *coverage*: four of the five realizations at each injection are bootstrap bin-resamples of a single re-centred dataset rather than independent noise realizations, and the per-realization recovered widths ($\sigma_\gamma \approx 0.16\text{--}0.17$) are narrower than the real-data marginal 0.382 of Eq. (9), as expected when the bin-to-bin scatter of an actual observation is not regenerated. The ± 0.382 that enters the z -distances of Table IV is therefore validated as a posterior width from the real likelihood, not by this injection; an independent-noise coverage test is left as future work.

D. The model’s own nHz prediction: a null

The official posterior and the refit above are the *measured* slope values; they are not by themselves a test of the matter bounce, since $\gamma = 3$ was not derived from this model’s own spectrum but borrowed from Papanikolaou [8]’s low- k tail, whose source carries a small-scale enhancement that §VC shows this model’s spectrum does not have. We therefore propagate this model’s own curvature spectrum, $\Delta_c^2(k) = A_s(k/k_*)^{n_s-1}$ with $A_s = 2.1 \times 10^{-9}$, $k_* = 0.05\text{ Mpc}^{-1}$, $n_s = 0.9649$ (the in-lab curvature spectrum, reproducibility statement below), through the scale-independent A2 bounce transfer and the standard radiation-era induced-GW kernel [10] (kernel validated against the published scale-invariant closed form, 0.822542 vs. 0.8222, before any lab spectrum was inserted). Every background at or above §VC’s $T_B \gtrsim 6 \times 10^9\text{--}6 \times 10^{10}\text{ GeV}$ puts the entire NANOGrav band at $k\eta_B \leq 3.8 \times 10^{-10}$ (the value at $T_B = 10^8\text{ GeV}$, 2.3×10^{-8} , already suffices), seven or more orders inside the transfer’s validity domain, so this result is not limited by transfer uncertainty.

The fitted slope over $f \in [2, 60]\text{ nHz}$ is

$$\begin{aligned} \gamma_{\text{pred}} &= 5.070 \quad (n_s = 0.9649), \\ \gamma_{\text{pred}} &= 5.000 \quad (n_s = 1 \text{ dust bracket}), \end{aligned} \quad (10)$$

reproducing the analytic self-similarity result $\gamma = 5 - 2(n_s - 1)$ to 1 part in 10^4 , with predicted amplitude $\Omega_{\text{GW}} h^2(f_{\text{yr}}) = 1.45 \times 10^{-23}$ (5.88×10^{-23} for the dust bracket) against NANOGrav’s power-law value $\Omega_{\text{GW}} h^2(f_{\text{yr}}) = 2.622 \times 10^{-8}$ at f_{yr} , using the free- γ posterior amplitude ($\log_{10} A \approx -14.19$, $A \approx 6.46 \times 10^{-15}$) paired consistently with the free- γ slope $\gamma_{\text{HD}} = 3.2$ compared against below (source file in the reproducibility

statement below). Two distinct failures against the borrowed $\gamma = 3$ (both derived from the same single fiducial power-law spectrum, so not statistically independent): the slope is a Gaussian-equivalent 5.1σ (4.9σ , dust bracket) z -distance from NANOGrav's measured $\gamma_{\text{HD}} = 3.2 \pm 0.6$ (a 5–95% interval width, consistent with the Savage–Dickey stance of Sec. IV), and the amplitude is a shortfall of $10^{15.2}$ ($10^{14.6}$, dust bracket) – fifteen orders of magnitude below the PTA band. The model predicts an induced-GW background far below current sensitivity and does not account for the NANOGrav signal: this channel is a **null**, not a supporting consistency.

The comparison above is the *second-order* (scalar-induced) background only; the model's own *first-order* (primordial) tensor background at nHz, propagated with the same $f \leftrightarrow k$ map and transfer prefactor and the same background's tensor tilt $n_T = n_s - 1$, dominates the induced background by roughly six decades at the CMB bound $r < 0.036$ [24] ($\Omega_{\text{GW}}^{(1)} h^2(f_{\text{yr}}) < 2.5 \times 10^{-17}$ against the induced 1.45×10^{-23}), and by 9.1 decades using the scheme-S1 $r_{\text{after}} = 24$ first-order tensor background of Sec. VII ($\Omega_{\text{GW}}^{(1)} h^2(f_{\text{yr}}) = 1.7 \times 10^{-14}$, propagated with the same kernel), which is already excluded at k_* by the same BICEP/Keck bound independently of any of the three channels. It remains $10^{6.2}$ below NANOGrav's 2.622×10^{-8} under this S1 value; under the selected S2 continuation of Sec. III A, $r_{\text{after}} \approx 9.4 \times 10^2$ scales this to $\Omega_{\text{GW}}^{(1)} h^2(f_{\text{yr}}) \approx 6.6 \times 10^{-13}$, narrowing the margin to $10^{4.6}$ – the null verdict is unchanged either way. Its slope, $\gamma_{\text{pred}}^{(1)} = 5.035$, is within 0.04 of the induced value, so Channel I is a null in both amplitude and slope whichever tensor order is compared (reproducibility statement below).

Reaching $\gamma = 3$ would require $P_{\mathcal{R}} \propto k^1$ sustained across the full 2–60 nHz band (the lab spectrum instead has $n_s - 1 = -0.0351$, or 0 in the dust bracket) and a $10^{7.2}$ amplitude enhancement at $k \approx 10^7 \text{ Mpc}^{-1}$ – precisely the PBH-forming regime §V C shows this model's spectrum falls seven decades short of. Alternatively, placing the required feature at the bounce scale itself needs $T_B \approx 2.3 \text{ GeV}$, 9.6 decades below §V C's own $T_B \gtrsim 6 \times 10^9$ – $6 \times 10^{10} \text{ GeV}$ condition. Channels I and II therefore assume mutually inconsistent spectra if $\gamma = 3$ is claimed, and the lab's own spectrum sides with Channel II's null. We retain the $\gamma = 3$ citation only as a flagged contrast case, requiring an enhancement this model does not produce, not as this model's prediction. Nor does this channel constrain f_{NL} even under that contrast case: it measures a slope, not a bispectrum amplitude, and would be identical at $-35/16$ and $-35/8$.

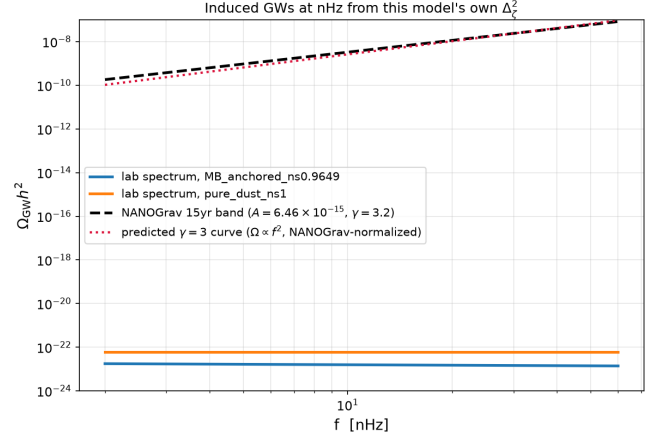


FIG. 1. Channel I null: $\Omega_{\text{GW}} h^2(f)$ induced by this model's own curvature spectrum (CMB-anchored $n_s = 0.9649$ and the $n_s = 1$ dust bracket), propagated through the validated Kohri–Terada radiation-era kernel [10] and the scale-independent A2 transfer, against NANOGrav's 15-yr power-law posterior central line ($\gamma_{\text{HD}} = 3.2 \pm 0.6$; the ± 0.6 is not shaded) and, for contrast, the literature $\gamma = 3$ curve ($\Omega_{\text{GW}} \propto f^2$, NANOGrav-normalized) this model does not produce. The predicted slope, $\gamma_{\text{pred}} = 5.07$ (5.00, dust), is a Gaussian-equivalent 5.1σ (4.9σ) z -distance from the measured slope; the predicted amplitude is $10^{15.2}$ below the NANOGrav band. f is the present-day observed frequency. Source data are listed in the reproducibility statement.

V. CHANNEL II: PRIMORDIAL BLACK HOLE ABUNDANCE

A. Press–Schechter with local quadratic non-Gaussianity: a truncated-map bound

Writing $\zeta = \zeta_G + A(\zeta_G^2 - \sigma^2)$ with $A = \frac{3}{5} f_{\text{NL}}$ and $\zeta_G \sim \mathcal{N}(0, \sigma^2)$ [13, 14], the mass fraction is $\beta(\zeta_c) = P(\zeta > \zeta_c)$, obtained exactly by inverting the quadratic. For $f_{\text{NL}} < 0$ this map is a downward parabola with an absolute ceiling

$$\zeta_{\text{max}} = -\frac{5}{12 f_{\text{NL}}} + \frac{3}{5} |f_{\text{NL}}| \sigma^2, \quad (11)$$

whose *leading* term scales as $1/|f_{\text{NL}}|$ and exactly doubles when $|f_{\text{NL}}|$ is halved; quoting that leading term alone gives $\zeta_{\text{max}} = 0.09524$ at $-35/8$ and 0.19048 at $-35/16$. The subleading $\frac{3}{5} |f_{\text{NL}}| \sigma^2$ term is dropped from those two numbers; restoring it at a representative $\sigma = 0.1$ gives the full $\zeta_{\text{max}} = 0.1215$ at $-35/8$ and 0.2036 at $-35/16$ (ratio 1.68, not exactly 2). At the standard curvature thresholds $\zeta_c \approx 0.45$ – 1 this ceiling forces $\beta \equiv 0$ identically for *both* candidate values, so a first pass restricted to the low-threshold regime ($\zeta_c \lesssim 0.15$) is the only place the quadratic map produces a nonzero answer at all – and there it reported $f_{\text{PBH}}(-35/16) > f_{\text{PBH}}(-35/8)$ by three to ten orders of magnitude, with the halved value landing inside the $10^{-3} \leq f_{\text{PBH}} \leq 1$ band reported by Choudhury *et al.* [16] for negative- f_{NL} bounce scenarios. That order-

ing was the sharpest statement the channel could make, but the ceiling that produced it is an artefact of truncating the local map at quadratic order, not a physical statement: it is exactly why the literature this comparison is drawn against uses the compaction-function formation criterion instead [16], and redoing the calculation on that criterion is the subject of the rest of this section.

B. The compaction-function result

Following Young, Byrnes & Sasaki (2019) [17] and Choudhury *et al.* (2025) [16], formation is set by the peak compaction $C(r_p) = \delta_p$ exceeding a threshold C_{th} , with the radiation-era compaction function $C = C_{\text{lin}} - C_{\text{lin}}^2/(4f(w))$, $f(w) = 2/3$, and $C_{\text{lin}} = JC_G$ carrying the same quadratic local map ($J = 1 + \frac{6}{5}f_{\text{NL}}\zeta_G$) applied to the linear compaction C_G rather than to ζ itself. Because C_G is an unbounded Gaussian variable, the map $\zeta_G \rightarrow C_{\text{lin}}$ has *no* ceiling, so every threshold $C_{\text{th}} < C_{\text{max}} = f(w) = 2/3$ is reachable at every f_{NL} : the Eq. (11) artefact is removed by construction. We integrate the joint Gaussian PDF of (C_G, ζ_G) over the type-I formation domain (Musco 2019 [18]; C_{th} scanned over $\{0.4, 0.5, 0.6\}$) with a critical-scaling mass function ($\gamma_{\text{crit}} = 0.36$, $K = 4$), using the Gaussian window for C_G and the spherical window (with radiation transfer function) for ζ_G , exactly as in Choudhury *et al.* Eqs. (30–66). The one ingredient of their calculation that is not reconstructible from the published paper is the regularized-renormalized-resummed one-loop curvature spectrum sourced by their EFT-of-bounce-plus-USR mode history: no closed-form $\Delta_\zeta^2(k)$, loop-counterterm normalisation, or EFT parameter set is printed. We use a lognormal stand-in spectrum, scanned over width $\Delta \in \{0.35, 0.5, 0.8\}$ and peak-scale $r_p k_p \in \{0.75, 1.0, 1.5\}$, and scan the amplitude A rather than adopting theirs.

At fixed curvature amplitude, calibrated so the Gaussian case gives $f_{\text{PBH}} = 1$, this criterion *reverses* the first-pass ordering: across the full 27-point $(\Delta, r_p k_p, C_{\text{th}})$ grid, $f_{\text{PBH}}(-35/16) < f_{\text{PBH}}(-35/8)$ at *every* point (Table V, Fig. 2), because the correlation $\gamma_{\text{cr}} \equiv \sigma_{\text{cr}}^2/(\sigma_c \sigma_r)$ – with σ_c^2 the compaction-function variance, σ_r^2 the curvature variance, and σ_{cr}^2 their cross-variance, all evaluated with the same real-space top-hat window at the peak scale r_p used for C_{th} – that governs the sign of the non-Gaussian correction is large and positive (0.766–0.968) at these spectrum shapes: the Gaussian-case formation channel rides a correlated ridge that negative f_{NL} destroys, forcing the system into an anti-correlated, probability-costly regime whose leverage *grows* with $|f_{\text{NL}}|$. $-35/8$ therefore always out-produces $-35/16$ here – the opposite of the truncated-quadratic-map result. The one analytic statement of the first pass, that Eq. (11)’s leading term exactly doubles when $|f_{\text{NL}}|$ is halved (the full expression, including the dropped σ^2 term, does not), remains arithmetically true of that leading term but is now physically empty: it described a truncation artefact that the com-

paction criterion does not share.

f_{PBH} itself is not a quotable constraint: because it depends exponentially on γ_{cr} , it moves by more than 100 decades across the grid at fixed target amplitude (Table V), tracking the unreproducible spectrum shape (the primordial spectrum’s unreconstructible detailed profile, discussed further in Sec. VB). What *is* robust is the ratio of curvature amplitudes needed to reach a fixed target abundance, since K , Ω_{DM} , g_* , and M_H all cancel in it: reaching the floor of the Choudhury *et al.* band, $f_{\text{PBH}} = 10^{-3}$, requires

$$\frac{A(-35/16)}{A(-35/8)} = 1.84 [1.76, 1.89], \quad \text{std} = 0.03, \quad n = 144, \quad (12)$$

restricted to $\gamma_{\text{cr}} \in [0.267, 0.630]$, the interval this model’s own near-scale-invariant spectrum shape occupies (on the narrower, original 27-point fiducial grid, $\gamma_{\text{cr}} \in [0.766, 0.968]$, the same ratio is 1.732 [1.610, 1.809], std = 0.050, and must not be quoted as universal), i.e. stable to $\pm 2\%$ – 6% while f_{PBH} itself moves by more than 100 dex. This ratio is the channel’s most reproducible output – it holds at every one of the 144 grid points in this model’s own γ_{cr} range and is robust to spectrum shape (std = 0.03 on a mean of 1.84; the original, narrower 27-point fiducial grid gives 1.732 ± 0.050 and must not be quoted as universal) – and it is a result within a stated regime, not a claim of general validity. *Regime of validity.* The result lives on the anti-correlated ($\gamma_{\text{cr}} > 0$, $J > 1$) branch identified above, and it holds at curvature excursions where the quadratic local map is not always perturbatively controlled: $1.2|f_{\text{NL}}|\sigma_r \approx 0.54$ – 1.01 at $-35/16$ and ≈ 1.09 – 2.02 at $-35/8$ across the grid – the same class of limitation Choudhury *et al.* report as their own $|f_{\text{NL}}| \lesssim 60$ perturbativity bound, and a limitation of the shared quadratic ansatz rather than of either calculation specifically. *Non-monotonicity.* At fixed curvature amplitude, $f_{\text{PBH}}(f_{\text{NL}})$ is strongly non-monotonic along the continuity path from the Gaussian case ($f_{\text{NL}} = 0$) to $-35/8$: it falls by ~ 55 decades to a minimum near $f_{\text{NL}} \approx -0.35$, then climbs back by ~ 53 decades through $-35/16$ to $-35/8$. Both candidate values lie past that minimum, on the branch where the anti-correlated ($\zeta_G < 0$) region of (C_G, ζ_G) space dominates formation and drives the ratio’s leverage; a reader inspecting only the two candidate points would not see this structure, so we record it here for transparency even though it does not change Eq. (12). The Gaussian-calibrated normalization amplitude at the representative grid point $(\Delta, r_p k_p, C_{\text{th}}) = (0.5, 1.0, 0.5)$ is $A_* = 0.131446$ (the amplitude at which $f_{\text{PBH}} = 1$ for $f_{\text{NL}} = 0$); Table V and Fig. 2 report f_{PBH} at this calibration. Reproducing Choudhury *et al.*’s reported band at $-35/8$ ($10^{-3} \leq f_{\text{PBH}} \leq 1$) is not forced by construction – the amplitude was fixed by the unrelated Gaussian-case normalisation – and it lands there for every γ_{cr} once the sign disagreement below $\gamma_{\text{cr}} \approx 0.85$ is traced to its source and resolved: the apparent *enhancement* there is an IR-divergent $O(\epsilon^2)$ artefact of the quadratic com-

TABLE V. Channel II. Compaction-function criterion: amplitude ratio $A(-35/16)/A(-35/8)$ required to reach $f_{\text{PBH}} = 10^{-3}$, and the *nominal (uncapped)* f_{PBH} columns – i.e. evaluated at the Gaussian-calibrated amplitude A_* with no upper cutoff applied – for representative points of the $(\Delta, r_p k_p, C_{\text{th}})$ grid (full 27-point grid in the reproducibility statement’s Channel II artifacts; the headline 1.84 ± 0.03 below comes from the 144-point in-coverage subset of the extended scan (reproducibility statement below), not from the 27-point grid). An extended 255-point scan over $\gamma_{\text{cr}} \in [0.196, 0.968]$ (two spectrum-shape families) gives 1.83 ± 0.05 overall (range $[1.61, 1.91]$); restricted to $\gamma_{\text{cr}} \in [0.267, 0.630]$, the interval this model’s own near-scale-invariant spectrum shape occupies, the ratio is **1.84 ± 0.03** (144 points, range $[1.76, 1.89]$; 78 lognormal and 66 power-law shapes whose γ_{cr} falls in that interval, the power-law family alone giving 1.839 ± 0.031); on the original 27-point fiducial grid ($\gamma_{\text{cr}} \in [0.766, 0.968]$) it is 1.732 ± 0.050 (range $[1.610, 1.809]$) and must not be quoted as universal. The ratio decreases with γ_{cr} at fixed spectrum family and C_{th} (ordinary-least-squares slope -0.20 per unit γ_{cr} over the 255-point scan and -0.20 over the 144 in-window points) and rises by $\simeq +0.03$ per $+0.1$ in C_{th} , so it is not monotonic across the union of families and thresholds. The abstract’s 1.84 ± 0.03 is therefore a shape-resolved scan result, not a union of a scan and an out-of-coverage point. Each row is evaluated at its own labeled $(\Delta, r_p k_p, C_{\text{th}})$ with its own Gaussian calibration; the per-row A_* are 0.1687, 0.1054, 0.1314, 0.3790, 0.0963 (reproducibility statement below). Any entry with $f_{\text{PBH}} > 1$ – including the $\gamma_{\text{cr}} \lesssim 0.8$ rows (*non-perturbative branch*; $1.2|f_{\text{NL}}|\sigma_r \approx 0.5$ – 2 , i.e. $O(\epsilon) \sim 1$ where the linearized criterion above is not controlled, IR-cutoff-dependent per the text) – is an *uncapped ratio* at that A_* , not a physical probability. The sign disagreement with Choudhury *et al.* is resolved above: negative f_{NL} suppresses the abundance, as they report (reproducibility statement below).

Δ	$r_p k_p$	C_{th}	γ_{cr}	$f_{\text{PBH}}(-35/16)$	$f_{\text{PBH}}(-35/8)$	ratio A
0.35	0.75	0.5	0.897	1.7×10^{-11}	1.0	1.767
0.35	1.5	0.5	0.968	5.3×10^{-107}	7.0×10^{-59}	1.673
0.50	1.0	0.5	0.888	3.6×10^{-14}	1.6×10^{-2}	1.734
0.80	0.75	0.6	0.766 [†]	9.5×10^5	6.6×10^9	1.809
0.80	1.5	0.4	0.857	1.9×10^{-14}	3.2×10^{-3}	1.677
mean ratio 1.732 [1.610, 1.809], std 0.050, $n = 27$						
[†] non-perturbative: $f_{\text{PBH}} > 1$ uncapped						

paction map’s saddle expansion ($\propto -(6\gamma_{\text{cr}}^2 - 1)\sigma_r^2$, positive only below $\gamma_{\text{cr}} = 1/\sqrt{6} \approx 0.408$ and formally divergent as the IR cutoff is removed), not a physical effect; the saddle expansion’s own flip point (0.408) undershoots the full numerical reversal (≈ 0.85), consistent with the perturbativity criterion above ($1.2|f_{\text{NL}}|\sigma_r \approx 0.54$ – 1.01) being approached over $0.408 \lesssim \gamma_{\text{cr}} \lesssim 0.85$, where higher orders in ϵ are not controlled; the IR-finite $O(\epsilon)$ term, $\epsilon = (6/5)f_{\text{NL}}\sigma_r$, suppresses at first order in ϵ at every γ_{cr} , so negative f_{NL} suppresses the compaction abundance at first order, and Choudhury *et al.* are right.

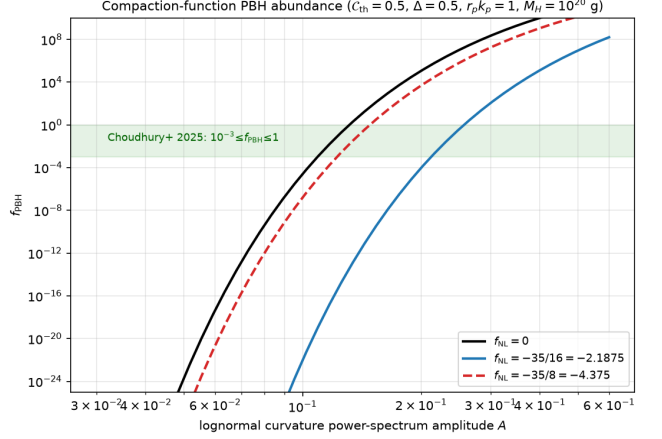


FIG. 2. Channel II, compaction-function criterion: f_{PBH} vs. curvature power-spectrum amplitude A at the representative grid point $(\Delta, r_p k_p, C_{\text{th}}) = (0.5, 1.0, 0.5)$, for the Gaussian case ($f_{\text{NL}} = 0$, calibrated to $f_{\text{PBH}} = 1$ at $A_* = 0.131446$) and the two candidate values; $-35/16$ requires higher A than $-35/8$ to reach the same f_{PBH} . The continuity scan underlying the non-monotonicity discussed in the text is reported in the Channel II artifacts of the reproducibility statement; the required amplitude-ratio grid (Eq. (12)) is in Table V.

Present-day abundance uses [15]

$$f_{\text{PBH}} = \frac{1}{\Omega_{\text{DM}}} \left(\frac{M_{\odot}}{M_H} \right)^{1/2} \left(\frac{g_*}{106.75} \right)^{3/4} \left(\frac{g_{*s}}{106.75} \right)^{-1} \times \frac{\beta_{\text{NG}}}{7.9 \times 10^{-10}}, \quad (13)$$

with $\Omega_{\text{DM}} = 0.674^1$, $g_* = g_{*s} = 106.75$, and a single horizon mass $M_H = 10^{20}$ g (matching the first pass, rather than the mass-integrated Eq. 66 of Choudhury *et al.*, an $O(1)$ width-factor deviation for a narrow peak). Full formalism, the itemized deviations from Choudhury *et al.*’s setup (mass-integration, width-factor, and spectrum-shape approximations), and validation are recorded in the compaction note cited in the reproducibility statement below.

¹ Quoted as printed in Choudhury *et al.*’s Eq. (66); it numerically coincides with the Planck 2018 reduced Hubble parameter h , not the dark-matter density parameter ($\Omega_{\text{DM}} \approx 0.264$). The required-amplitude *ratio* (Eq. (12)) and this section’s null verdict are unaffected, since Ω_{DM} cancels identically there; the *tabulated* f_{PBH} column and Fig. 2 are calibrated so the Gaussian case gives $f_{\text{PBH}} = 1$ at this quoted value and would shift by the factor $0.674/0.264 \approx 2.55$ under the dark-matter-density value (since $f_{\text{PBH}} \propto 1/\Omega_{\text{DM}}$) – they are not independent of which Ω_{DM} is used, though this $O(1)$ rescaling changes none of this section’s conclusions: the model’s own predicted f_{PBH} is exactly zero in double precision regardless (Sec. V C), seven orders of magnitude below either normalization.

C. The lab's own primordial spectrum: a null result

Sec. VB's spectrum shape is a lognormal stand-in with a *scanned* amplitude, because Choudhury *et al.*'s spectrum is not reconstructible from their published paper. Removing that dependence, Eq. (12)'s amplitude ratio can instead be evaluated for the spectrum this matter-contraction model actually predicts: a near-scale-invariant power law, $n_s - 1 = 12w/(1 + 3w)$, anchored to the Planck 2018 amplitude $A_s = 2.1 \times 10^{-9}$ at $k_* = 0.05 \text{ Mpc}^{-1}$ and tilted by the small deviation from $w = 0$ ($n_s = 0.9649$), extrapolated across the 10–15 decades to PBH scales under two assumptions: the equation of state does not evolve away from its CMB-calibrated value, and the bounce transfer is scale-independent, which Sec. III A establishes only for $k\eta_B \lesssim 10^{-2}$: satisfied at every PBH mass scale considered here only if the bounce energy is high enough. Requiring $k\eta_B \lesssim 10^{-2}$ at the smallest PBH mass used here (10^{15} g , $k \sim 10^{15}\text{--}10^{16} \text{ Mpc}^{-1}$) needs $a_B H_B \gtrsim 10^{17}\text{--}10^{18} \text{ Mpc}^{-1}$, i.e. a bounce temperature $T_B \gtrsim 6 \times 10^9\text{--}6 \times 10^{10} \text{ GeV}$ (using $k_B \simeq 1.71 \times 10^7 T_B [\text{GeV}] \text{ Mpc}^{-1}$, the in-lab spectrum) – thirteen to fourteen decades above the BBN scale ($\sim 1 \text{ MeV}$); a scale-independent transfer, $T_{\text{fNL}} \lesssim 1$ on any scheme considered here (S1's $< 1/2$ or S2's ≈ 1.03 on Quintin-type), cannot by itself close the 7-decade amplitude shortfall found below, so this condition, while formally required, does not affect the null verdict. On this shape, inside the model's own $\gamma_{\text{cr}} \in [0.267, 0.630]$ coverage the required-amplitude ratio is 1.84 ± 0.03 across a threshold scan $C_{\text{th}} \in \{0.4, 0.5, 0.6\}$: the ratio remains the channel's robust, spectrum-shape-stable output, but the narrower 1.732 ± 0.050 from the original fiducial grid must not be quoted as if it were universal.

Fed into the same compaction machinery unmodified, the lab's own spectrum delivers $\Delta_\zeta^2 \sim 5 \times 10^{-10}\text{--}1.3 \times 10^{-9}$ at PBH scales against a required 0.006–0.019 (the compaction threshold): a shortfall of 6.7–7.0 orders of magnitude in curvature amplitude ($\log_{10}$ ratio 6.75 at $-35/8$, 7.02 at $-35/16$), at every mass from 10^{15} g to $10^4 M_\odot$, over the same threshold scan and a four-decade IR-cutoff scan. f_{PBH} is exactly zero in double precision at every point (Gaussian-limit $\log_{10} \beta \approx -1.7 \times 10^9$ at $M_H = 10^{20} \text{ g}$, i.e. an $n_\sigma = 89,149$ Gaussian excursion; $89149^2/(2 \ln 10) = 1.73 \times 10^9$; the in-lab spectrum). **The PBH channel is therefore a null for this model's own predicted spectrum:** it neither constrains nor is evidenced by primordial black holes, and any quotable PBH signal would require a small-scale amplification mechanism the model does not contain – a survey of four such candidate mechanisms [3, 7] finds none applies within the model's validated ($k\eta_B \lesssim 10^{-2}$) domain, and the survey is supplemented by a direct evaluation: at $k\eta_B \approx 3$ (an illustrative bounce energy below §VC's validity condition, $T_B = 10^8 \text{ GeV}$, $k_B \approx 1.714 \times 10^{15} \text{ Mpc}^{-1}$, so $k \approx 5.1 \times 10^{15} \text{ Mpc}^{-1}$, inside this paper's own PBH band), the extreme small-

scale end of the PBH band, the scheme-S1 bispectrum is evaluated directly on exact (non-super-Hubble) LQC-dust mode functions (Sec. III) and gives $|f_{\text{NL}}| \approx 0.9\text{--}1.9$, which *worsens* rather than relieves the required Gaussian excursion (the required-excursion threshold, at the $M_H \sim 10^{15}\text{--}10^{16} \text{ g}$ end where the 7.0-decade Δ_ζ^2 deficit above applies – a $10^{3.5} = 3162$ Gaussian- σ excursion – moves to 3364σ); even the largest number produced anywhere in that scan (equilateral configuration, $|f_{\text{NL}}| \approx 1.2 \times 10^3$) leaves the threshold at 408σ against the 7.0-decade amplitude deficit established here, so no configuration, initial state, or evaluation time tested reopens this channel.

The same spectrum is checked against the FIRAS μ -distortion bound ($|\mu| < 9 \times 10^{-5}$): the lab's own prediction gives $\mu = 1.65 \times 10^{-8}$, safely allowed at 1.8×10^{-4} of the bound. For the early-supermassive-black-hole-seed channel (a possible use of PBH formation this ratio was originally motivated by), the *required* curvature amplitude for $10^3\text{--}10^6 M_\odot$ seeds is FIRAS-excluded by three orders of magnitude if broadband, and remains excluded for seeds $\geq 10^4 M_\odot$ even as a narrow ($\Delta = 0.5$) spectral peak; only the lightest ($10^3 M_\odot$) seeds evade the bound with a sufficiently narrow peak. Combined with the null above, the honest statement is two-part: the required SMBH-seed amplitude is excluded by three orders of magnitude (broadband, and for $\geq 10^4 M_\odot$ narrow-peak seeds) independent of this model, and this model's own spectrum falls far short of supplying it either way. Full derivation, the four-mechanism literature survey, and every number above are archived in the lab-spectrum script cited in the reproducibility statement below.

VI. CHANNEL III: THE LARGE-SCALE-STRUCTURE BISPECTRUM

A. Current constraints

The DESI DR1 analysis of 1,631,716 luminous red galaxies ($0.6 < z < 1.1$) and 1,189,129 quasars ($0.8 < z < 3.1$) [19] gives $f_{\text{NL}}^{\text{local}} = -3.6_{-9.1}^{+9.0}$ (asymmetric 9.0/9.1) at 68% confidence with a merger model for the quasar response, or $f_{\text{NL}}^{\text{local}} = +3.5_{-7.4}^{+10.7}$ (asymmetric 10.7/7.4) assuming universality. These two numbers come from mutually exclusive theoretical priors on the same DESI DR1 data (merger-response vs. universality) and are not directly comparable to each other; each is quoted here against its own prior. DESI DR1 constrains the observed, post-bounce local amplitude directly, so the transmitted value $f_{\text{NL}}^{\text{after}} = -1.25$ (Sec. III A) is what it should be compared against; this section also quotes the pre-bounce amplitude $f_{\text{NL}}^{\text{local}} = -35/16$ as the scale at which the contraction-phase prediction is defined, using the upper side of the merger-response error (+9.0, since the prediction lies above the central value) and the lower side of the universality error (−7.4, for the same reason); the full transmitted range is compared against

forecast reach below (Sec. VI, “Forecast reach”). At $f_{\text{NL}}^{\text{after}} = -1.25$ the prediction sits 0.26σ from the merger-response central value and 0.64σ from the universality one. The pre-bounce prediction sits 0.16σ from the former and 0.77σ from the latter. The discriminating power is $|f_{\text{NL}}|/\sigma = 0.24$: DESI DR1 is consistent with the prediction under either prior and cannot test it.

As an independent check of the pipeline these forecasts rest on, we re-analyze the public DESI DR1 QSO sample using DESI’s own published window matrix, full-randoms P_ℓ , and EZmock covariance ($z = 0.8\text{--}2.1$, $k \leq 0.08 h \text{Mpc}^{-1}$, 46 dof; b_1 and f_{NL} free, n_{shot} fixed at 0), and recover $f_{\text{NL}}^{\text{local}} = -2.2 \pm 25$ at a quasar response parameter $p = 1.6$ (not a p -value) $- 0.06\sigma$ from DESI’s published $-3.6^{+9.0}_{-9.1}$ (a reproduction under a different procedure than DESI’s own, not directly comparable to it – see the disclosed scope below), which reproduces the published constraint’s sign and scale while showing that DR1’s constraining power at this configuration is an order of magnitude too weak to separate $-35/16$ from $-35/8$, from zero, or from the published central value; the near-coincidence of the recovered central value with $-35/16$ is a coincidence, not evidence. This reproduction’s $\sigma = 25$ is $\sim 2.8\times$ the published 9.0 because wide-angle corrections (`PowerSpectrumOddWideAngleMatrix`) are not applied and only 2 of 5 imaging-systematics splits were run; full manifest cited in the reproducibility statement below.

B. Forecast reach

As established in Sec. III, the $k\eta_B$ window is satisfied most easily at this channel’s CMB/LSS pivot scale. On the Quintin-type background the selected scheme S2 (Sec. III A) gives the single value $f_{\text{NL}}^{\text{after}} = -1.25$ ($-35/16$); on the LQC and poly backgrounds the Bardeen continuation’s linear transfer alone gives -1.094 and -1.140 respectively, with the cubic term uncomputed there so the full $f_{\text{NL}}^{\text{after}}$ is only a bracket, $[-1.20, -0.09]$ and $[-1.27, -0.13]$ (the superseded S1 values, retained for comparison, were $\in [-0.65, -0.55]$); the S1 range across all three backgrounds is $[-1.20, -0.86]$ ($-35/8$) (Table III). These are this paper’s observable predictions for Channel III, not the pre-bounce contraction-phase amplitude. Table VI reports the bare significance $f_{\text{NL}}^{\text{after}}/\sigma_{f_{\text{NL}}}$ across the three backgrounds of Sec. III, with the pre-bounce amplitude’s bare significance retained as a clearly-labelled secondary row (i.e. assuming $T_{f_{\text{NL}}} = 1$, the transfer-free case this paper does not otherwise support). The matter-bounce bispectrum is also not exactly the local shape; a noise-weighted shape overlap $r_{\text{sh}} < 1$, which has not been re-derived at the $f_{\text{NL}}^{\text{after}}$ fiducial used in this paper – so no numeral is quoted here – would let an experiment quoting σ_{local} constrain the bounce amplitude at $\sigma_{\text{local}}/r_{\text{sh}}$; the r_{sh} -projected numbers are omitted pending that re-derivation.

At the transmitted amplitude the two candidate values

TABLE VI. Channel III. Bare detection significance $|f_{\text{NL}}^{\text{after}}|/\sigma_{f_{\text{NL}}}$ of the transmitted amplitude across the three backgrounds of Sec. III (range shown; Table III gives per-background values). The pre-bounce row (assuming $T_{f_{\text{NL}}} = 1$) is a secondary reference, not this paper’s prediction. The scheme-S2 row, selected on the Quintin-type background by the Bardeen-potential argument of Sec. III A, is this paper’s Channel III prediction there, superseding the S1 row for Quintin-type; the S1 rows are superseded on the LQC and poly backgrounds too by a Bardeen continuation (Sec. III A, Table III), whose cubic term is not computed there, so no separate significance row is given for them. No $-35/8$ value is computed for S2 (not scaled from $-35/16$; see text). Forecast σ values are quoted from the cited abstracts. MegaMapper is an unapproved design and is quoted only at its published order-unity level; no systematic budget is transferred to it. Every tabulated significance is an *upper bound*: it assumes a shape overlap $r_{\text{sh}} = 1$ with the local template, and the matter-bounce shape’s true overlap $r_{\text{sh}} < 1$ (not yet re-derived at this paper’s $f_{\text{NL}}^{\text{after}}$ fiducial) can only reduce these numbers. Values re-derived by the Channel III artifacts of the reproducibility statement.

Survey	$\sigma_{f_{\text{NL}}}$	bare ($-35/16$)	bare ($-35/8$)
<i>scheme S1, all three backgrounds:</i>			
SPHEREx, bispectrum only [20]	0.7	0.7–0.9 σ	1.2–1.7 σ
SPHEREx, $P + B$ target [20, 21]	0.5	1.0–1.3 σ	1.7–2.4 σ
MegaMapper class [22, 23]	~ 1	0.50–0.65 σ	0.86–1.20 σ
<i>scheme S2 (selected), Quintin-type only ($f_{\text{NL}}^{\text{after}} \approx -1.24$ to -1.25):</i>			
SPHEREx, bispectrum only	0.7	1.78 σ	not computed
SPHEREx, $P + B$ target	0.5	2.49 σ	not computed
MegaMapper class	~ 1	1.25 σ	not computed
<i>pre-bounce secondary (assuming $T_{f_{\text{NL}}} = 1$):</i>			
SPHEREx, bispectrum only	0.7	3.13 σ	6.25 σ
SPHEREx, $P + B$ target	0.5	4.38 σ	8.75 σ
MegaMapper class	~ 1	2.19 σ	4.38 σ

separate by $|\Delta f_{\text{NL}}^{\text{after}}| \approx 0.35\text{--}0.55$ against $\sigma_{f_{\text{NL}}} = 0.5\text{--}0.7$ at SPHEREx sensitivity: $0.5\text{--}1.1\sigma$, so this channel does not discriminate $-35/16$ from $-35/8$ at current forecast reach once scheme-S1 transmission is applied; it remains the only channel where the factor of two would be observationally consequential *if* the pre-bounce amplitude survived transmission unsuppressed, which Sec. III shows it does not.

VII. THE TENSOR AMPLITUDE OF THE MODELLED BACKGROUND

Every result above is conditional on the canonical, constant-sound-speed dust contraction used for the amplitude and its transmission. That background is not free of its own tensor spectrum. In the dressed-metric scheme the tensor and scalar canonical variables obey the identical equation $\mu'' + (k^2 - a''/a)\mu = 0$ from the identical Bunch–Davies vacuum, so $r = 16\epsilon$ identically for this class of background, and the dust contraction’s $\epsilon = 3/2$ fixes $r = 24$. The tensor mode

obeys $h'' + 2(a'/a)h' + k^2h = 0$ with $c_T = 1$: it contains no $1/H$, no ϵ , no c_s , and none of the scalar constraint variables, so the S1/S2 continuation ambiguity – a choice of scalar variable z through $H = 0$ – cannot act on it, and because the S1 scalar equation with $z = a$ is literally the tensor equation, the tensor transfer λ_T equals the S1 scalar transfer $\lambda_\zeta^{\text{S1}}$ *identically*, verified to 1.4×10^{-14} on the Quintin-type background at $k\eta_B = 10^{-3}, 3 \times 10^{-3}, 10^{-2}$ by an independent tensor-equation solver, and to 8.5×10^{-9} on the poly and LQC backgrounds (Sec. III). So $r_{\text{after}} = r_{\text{before}}(\lambda_T/\lambda_\zeta)^2$ depends on the scheme only through the scalar denominator: $r_{\text{after}}^{\text{S1}} = 24.0$ is an identity, unchanged by the bounce, exceeding the BICEP/Keck bound $r < 0.036$ [24] by $\sim 670\times$; in S2 – the scheme selected on the Quintin-type background by the Bardeen-potential argument of Sec. III A – the scalar transfer under the raw-ADM continuation is $\lambda_\zeta^{\text{S2}} = 0.970$ against the shared tensor transfer $\lambda_T = 6.06$, so the bounce *amplifies* r by $(\lambda_T/\lambda_\zeta^{\text{S2}})^2 \approx 39\times$ to $r_{\text{after}}^{\text{S2}} \approx 9.4 \times 10^2$, $\sim 2.6 \times 10^4\times$ above the bound – this paper’s selected value on this background, strengthening rather than curing the tensor problem found in S1. The same construction now extends to the LQC and poly backgrounds, whose $\dot{H} = 0$ crossings are smooth rather than a jump (Sec. III A): with the tensor transfer identical to $\lambda_\zeta^{\text{S1}}$ there too and the Bardeen-continuation scalar transfer a fraction $R = 1/2$ (LQC) and $R = 3/8$ (poly, principal-value) of it, $r_{\text{after}} = 24/R^2$ gives $r_{\text{after}} \approx 96.0$ (LQC) and 170.6 (poly) – $2.7 \times 10^3\times$ and $4.7 \times 10^3\times$ the BICEP/Keck bound, against $6.7 \times 10^2\times$ under S1’s common $r_{\text{after}} = 24$. On every background evaluated, including the two where the scalar continuation carries an $O(1)$ prescription freedom (Sec. III A), both continuations exceed the bound by more than two orders of magnitude, so neither of the two schemes evaluated here rescues any of them from the long-known matter-bounce large- r problem [2, 3], applied here to the lab’s own backgrounds rather than assumed from the literature. Similarly, the exact dust contraction gives $n_s = 1$; the observed $n_s = 0.9649$ is an anchor fixing the contraction’s equation of state to $w = -0.0029$ via $n_s - 1 = 12w/(1 + 3w)$, not a prediction of the model. The model’s one genuinely predictive tilt statement is the consistency relation $n_T = n_s - 1 = -0.035$, sharply different from single-field inflation’s $n_T = -r/8$, and a discriminator that is falsifiable in principle and survives everything else in this section – though at tensor-viable r no planned experiment reaches $\sigma(n_T) \sim 0.03$, so this is a statement about the model’s structure, not a near-term test. Propagating the scheme-S1 $r_{\text{after}} = 24$ first-order tensor background to nHz through the same kernel used in Sec. IV D gives $\Omega_{\text{GW}}h^2(f_{\text{yr}}) = 1.7 \times 10^{-14}$, still $10^{6.2}$ below NANOGrav’s 2.622×10^{-8} (the same free- γ posterior amplitude used throughout Sec. IV D); under the selected S2 continuation this narrows to $10^{4.6}$ (Sec. IV D’s footnote): either way, the PTA null of Sec. IV D is unaffected by this section’s own tensor amplitude. A tensor-sense $r = 0.84$ is

not used here: it traces to the noise-weighted bispectrum shape-overlap coefficient r_{sh} of Sec. VI, a shape correlation and not a tensor-to-scalar ratio.

VIII. THE JOINT (r, f_{NL}) NO-GO AND THE CURVATON ROUTE

The textbook remedy for a large tensor-to-scalar ratio in a contracting background is a reduced constant scalar sound speed $c_s < 1$, which suppresses r while enhancing the scalar bispectrum. For the dust contraction with constant c_s , two independent symbolic routes (a general- ν Hankel limit and the exact $q = 2$ mode functions) give

$$r = 16\epsilon c_s^{2\nu-2} = 24 c_s, \quad (14)$$

reproducing Li, Quintin, Wang & Cai (2016) [4] Eq. (3.18) and the $r = 24$ of Sec. VII at $c_s = 1$. The squeezed local amplitude follows from the same cubic action (their Eq. 4.18), whose ζ^3 vertex carries the $P_{,XXX}$ coefficient $\lambda \equiv X^2 P_{,XX} + \frac{2}{3} X^3 P_{,XXX}$ explicitly. Writing $\Lambda \equiv \lambda/\Sigma$, the general- λ amplitude is

$$f_{\text{NL}}^{\text{pre}}(c_s, \Lambda) = -\frac{245}{16} + \frac{105}{8c_s^2} - 30\Lambda, \quad (15)$$

which at $\Lambda = 0$, $c_s \rightarrow 1$ reproduces this paper’s own from-scratch $-35/16$ exactly (Sec. II). Li *et al.*’s quoted form is the specialization of Eq. (15) to the $P \propto X^n$ kinetic sector, $\Lambda = (1 - c_s^2)/(6c_s^2)$ (their Eq. A.20):

$$f_{\text{NL}}^{\text{pre}}(c_s)|_{P \propto X^n} = -\frac{165}{16} + \frac{65}{8c_s^2}, \quad (16)$$

their Eq. (5.1). Eq. (16) is the line tabulated in the window below; every general- Λ number quoted later in this section is measured from the $\Lambda = 0$ baseline of Eq. (15), not from Eq. (16). The bounce’s own cubic contribution, evaluated at the same constant c_s (scheme S1, $z = a$, $\epsilon_{\text{eff}} = 1/2$), generalizes Eq. (7) to

$$\Delta f_{\text{NL}}^{\text{bounce}}(c_s) = -\frac{5}{24} \rho_B \frac{6c_s^2 - 5}{c_s^4}, \quad (17)$$

reproduced by the full numerical in-in integral to 3×10^{-4} and reducing to Eq. (7) at $c_s = 1$ to 4×10^{-6} (reproducibility statement below); the c_s dependence enters the mode functions only through the sound horizon ($c_s^2 k^2$ in the Mukhanov–Sasaki equation, with the Bunch–Davies Wronskian already carrying $1/\sqrt{2c_s k}$), so the whole effect is carried by the vertex coefficients, dominated (99.97%) by $\zeta \dot{\zeta}^2$. Eq. (17) changes sign at $c_s = \sqrt{5/6} = 0.913$ and diverges as $1/c_s^4$, faster than the transmitted contraction term $T f_{\text{NL}}^{\text{pre}} \propto 1/c_s^2$, so $f_{\text{NL}}^{\text{pre}}(c_s) = T f_{\text{NL}}^{\text{pre}}(c_s) + \Delta f_{\text{NL}}^{\text{bounce}}(c_s)$ is used below rather than the transfer term alone. This evaluation holds for the full $P(X, \phi)$ cubic action, including a nonzero $P_{,XXX}$ coefficient $\lambda \equiv X^2 P_{,XX} + (2/3) X^3 P_{,XXX}$: λ enters the quadratic action

TABLE VII. The joint (r, f_{NL}) window in c_s , with the bounce’s own c_s -dependent cubic term (Eq. (17)) included at the same c_s (Quintin-type background, $T_{f_{\text{NL}}} = 0.165$; ranges span the three backgrounds of Table III; reproducibility statement below). The tensor requirement and every f_{NL} requirement are mutually exclusive, and the exclusion does not rest on the deep small- c_s extrapolation: the tree-level calculation itself expires at $c_s \simeq 0.08$, where r is already $\simeq 49\times$ the BICEP/Keck bound.

requirement	c_s	implied r
$r < 0.036$ (BICEP/Keck)	$< 1.5 \times 10^{-3}$	0.036
tree-level control ^a	$\gtrsim 0.073\text{--}0.079$	$\gtrsim 1.8$
$f_{\text{NL}}^{\text{after}} \in [-11.1, 9.3]$ (Planck 95%) ^b	≥ 0.525	≥ 12.6
$f_{\text{NL}}^{\text{after}} \in [-6.0, 4.2]$ (Planck 68%) ^b	≥ 0.624	≥ 15.0

^a Criterion 1.2 $|f_{\text{NL}}^{\text{after}}| \zeta_{\text{rms}} \leq 1$ with $\zeta_{\text{rms}} = \sqrt{A_s}$, as in Sec. V B.

^b Planck 2018 local constraint [28].

nowhere ($\partial c_s^2 / \partial P_{,XXX} = \partial \Sigma / \partial P_{,XXX} = 0$, so $r = 24c_s$ is exactly λ -independent), and its ζ^3 vertex in the bounce window is odd in conformal time about the symmetric bounce and cancels identically in the S1 in-in integral (verified to 2.8×10^{-7} across $\lambda/\Sigma \in [-1, 1]$, the matter-contraction line, and the DBI line; reproducibility statement below), so Eq. (17) also holds for any λ . $f_{\text{NL}}^{\text{pre}}(c_s)$ on the $P \propto X^n$ line of Eq. (16) (not the $\Lambda = 0$ baseline of Eq. (15), whose own root is $c_s = \sqrt{6/7} = 0.926$) itself changes sign at $c_s = 0.8876$ (negative above, positive below): on $c_s \in [0.444, 0.888]$, the branch that is otherwise CMB-viable by $|f_{\text{NL}}^{\text{after}}|$, the k -essence prediction is a *positive* local amplitude, so the flagship negative sign survives only for $c_s \gtrsim 0.89$. The transfer through the bounce is c_s -independent to $\lesssim 5 \times 10^{-11}$ across all three backgrounds, because lowering c_s only rescales the mode’s effective wavenumber and the transfer is already flat over $k\eta_B \ll 1$.

Including the bounce’s own term moves the Planck-viable boundary (upper edge binding, since $f_{\text{NL}}^{\text{after}} > 0$ throughout this window) from $c_s \geq 0.477$ ($r \geq 11.4$) without it to $c_s \geq 0.624/0.632/0.627$ (Quintin/LQC/poly, $r \geq 15.0/15.2/15.0$) against the 68% interval $[-6.0, +4.2]$, and to $c_s \geq 0.525$ ($r \geq 12.6$) against the 95% interval $[-11.1, +9.3]$ – nearly background-independent because the bounce term $\propto \rho_B$ and the transmission $(1 - \rho_B)/2$ are anti-correlated.

Where the calculation stops. Toward small c_s the bounce term grows as $1/c_s^4$, and the tree-level in-in bispectrum with a quadratic local map ceases to be controlled long before the tensor-viable sound speed is reached. Applying this paper’s own perturbativity criterion of Sec. V B, $1.2|f_{\text{NL}}^{\text{after}}|\zeta_{\text{rms}} \leq 1$ with $\zeta_{\text{rms}} = \sqrt{A_s} = 4.6 \times 10^{-5}$, gives $|f_{\text{NL}}^{\text{after}}| \leq 1.8 \times 10^4$ and hence a floor $c_s \gtrsim 0.073\text{--}0.079$ across the three backgrounds (reproducibility statement below). At $c_s = 1.5 \times 10^{-3}$ the formal value of $f_{\text{NL}}^{\text{after}}$ is $\sim 10^{11}$, but that is 6.9 decades past the criterion and is reported here only as the for-

mal continuation of Eqs. (16) and (17), *not* as a prediction; the $1/c_s^4$ law also rests on the S1 prescription ($z = a$, $\epsilon_{\text{eff}} = 1/2$), which is an assumption of the scheme and has no independent support at small c_s . Nothing in the no-go depends on it. At the perturbative floor itself $r \gtrsim 1.8$, already $\simeq 49\times$ the BICEP/Keck bound, so the tensor channel alone closes the window over the entire range in which the non-Gaussian calculation is meaningful, and the only quantitatively controlled edge of the f_{NL} window is the Planck one at $c_s \geq 0.525$ (95%). The two regions are disjoint by a factor ~ 350 in c_s at 95% (BICEP/Keck-viable $c_s < 1.5 \times 10^{-3}$ against Planck-viable $c_s \geq 0.525$), rising to ~ 416 at 68%: no constant- c_s value keeps both the tensor ratio and the local non-Gaussianity within current bounds, and the no-go is strengthened, not relaxed, by evaluating both terms at the same c_s . The 95% figure is the one quoted in the abstract. Eq. (17)’s own sign flip at $c_s = 0.913$ sits close to $f_{\text{NL}}^{\text{pre}}$ ’s at $c_s = 0.888$ above, so both terms change sign within the same narrow window and the flagship negative sign survives only above it. This independently reproduces and strengthens Li *et al.*’s (2016) own no-go statement (their $r < 0.07$ era bound gives $c_s \lesssim 0.0029$; the tighter current $r < 0.036$ moves this to $c_s < 1.5 \times 10^{-3}$ and $f_{\text{NL}}^{\text{pre}} = 3.6 \times 10^6$, a factor 3.8 tighter). The alternative route of Quintin *et al.* (2015) [3] – a scalar-only super-Hubble amplification through the bounce rather than a sound-speed change – also fails on the three backgrounds used throughout this paper: their Eq. (31) requires an amplification $\lambda \gtrsim 49$ for $r < 0.12$, while the velocity-dip evaluation of Sec. III (Eq. 79 of that reference, evaluated on these backgrounds) finds $\lambda = 4.0\text{--}6.1$ with no net r suppression. This entire c_s -window analysis is carried out in scheme S1, where r and f_{NL} are computed on the same transmission; the alternative-route test of Quintin *et al.* above is likewise available only in S1. Scheme S2 – now the selected continuation on the Quintin-type background (Sec. III A) – gives a tensor ratio larger still ($r_{\text{after}}^{\text{S2}} \approx 9.4 \times 10^2$, Sec. VII), so the joint exclusion is not relieved – if anything it is strengthened – by moving to S2. Restoring the cubic coefficient λ does not open the window either. Measuring from the $\Lambda = 0$ baseline of Eq. (15) – not from Eq. (16), which already carries Λ at its $P \propto X^n$ value – a nonzero Λ adds the purely local-shaped ($\propto \sum_i k_i^3$) term -30Λ to $f_{\text{NL}}^{\text{pre}}$ (Li *et al.* Eq. 4.18) and leaves $r = 24c_s$ untouched (Sec. VIII); note that on both physical lines $\Lambda \propto 1/c_s^2$, so the shift is constant in *shape*, not in c_s . Scanning $\Lambda \in [-1, 1]$ together with the matter-contraction line $\Lambda = (1 - c_s^2)/(6c_s^2)$ and the DBI line $\Lambda = (1 - c_s^2)/(2c_s^2)$ – under the same Planck 95% criterion $|f_{\text{NL}}^{\text{after}}| \leq 9.3$ as Table VII and across all three backgrounds of Table III – the best case is the DBI line on the LQC-effective-dust background, giving $r_{\text{min}} = 10.3$ at $c_s = 0.430$ ($286\times$ BICEP/Keck); on the 68% edge $|f_{\text{NL}}^{\text{after}}| \leq 4.2$ it is $r_{\text{min}} = 11.9$ at $c_s = 0.494$, and the best constant Λ reaches only $r_{\text{min}} = 11.7$ at 95%. Every window number in this section carries the criterion and background set it was computed under (repro-

ducibility statement below). Cancelling the $1/c_s^2$ growth of Eq. (15) outright would require $\Lambda = 7/(16c_s^2)$, which by Li *et al.*'s Eq. (A.19) [4] corresponds to a sound-speed running $s \equiv \dot{c}_s/(Hc_s) = 39/16$, far outside the $|s| \ll 1$ regime under which the amplitude being cancelled is derived. *The single-field matter bounce, for the full $P(X)$ k -essence class at constant sound speed ($s = 0$), is therefore excluded jointly by r and f_{NL} on the backgrounds and channels used in this paper, independently of the cubic coefficient λ ; a time-dependent $c_s(\eta)$ lies outside the present computation, which evaluates every vertex at fixed c_s (reproducibility statement below). The exact amplitude $-35/16$ and its transmitted values remain correct statements about this background; the background itself is excluded on tensor grounds independent of f_{NL} , and no sound-speed choice rescues it.*

A route that survives is a light curvaton-type spectator riding the same contraction. Because it obeys the same Mukhanov–Sasaki operator as the adiabatic mode at constant ϵ , it inherits the identical tilt $n_s - 1 = 12w/(1+3w)$, and for a quadratic curvaton with sudden decay fraction r_{dec} ,

$$r = \frac{24}{1 + \frac{4}{3}r_{\text{dec}}^2(M_{\text{pl}}/\sigma_*)^2}, \quad f_{\text{NL}}^{\text{curv}} = \frac{5}{4r_{\text{dec}}} - \frac{5}{3} - \frac{5r_{\text{dec}}}{6}, \quad (18)$$

(Lyth–Ungarelli–Wands [26]; Sasaki–Valiviita–Wands [27], applied here to Cai, Brandenberger & Zhang's 2011 [25] matter-bounce curvaton), which is (r, n_s) -tunable to observational viability and gives an $O(1)$ non-Gaussianity in its own right (+9.30 to -1.25 for $r_{\text{dec}} \in [0.113, 1]$, negative for $r_{\text{dec}} > 0.581$). The contraction's own intrinsic $-35/16$ does not disappear in this picture; it enters the observable bispectrum suppressed by the curvaton weight $(r/24)^2 \simeq 2 \times 10^{-6}$ at $r = 0.036$. Taking a 1σ SPHEREx-class threshold $\sigma_{f_{\text{NL}}} \simeq 0.5$ as the detectability criterion, the diluted intrinsic term clears it only for $r > 11.5$ before the bounce transfer, or $r > 21$ – 24 after it on the S1 band of Table III ($r > 15.2$ at the selected S2 value -1.25) – the threshold on r moves, not r itself, which is unchanged through the bounce in S1 (Sec. VII). Both values are far above the tensor-viable $r < 0.036$ and bracket the single-field $r = 24$ this section excludes, so the intrinsic amplitude is unobservable on this route; the curvaton's own $O(1)$ f_{NL} would dominate any signal. Cai, Brandenberger & Zhang's [25] own quoted $-320/\pi^4 \approx -3.29$ is arithmetic on their order-of-magnitude estimate, not an independently derived result, and is not adopted here. Converting this route into a concrete, checkable prediction requires an entropy-sector kinetic amplification mechanism and a demonstration that σ_* stays constant across the bounce backgrounds used in this paper; neither is modelled here (Sec. IX C).

IX. DISCUSSION

A. The joint statement

The matter-bounce prediction $f_{\text{NL}}^{\text{local}} = -35/16$ – independently re-derived here within the in-in formalism, confirming Li *et al.* (2016) and localizing the factor-of-two slip in Cai *et al.* (2009), and now confirmed by a method-independent route (Sec. IID) – is a correct statement about a background that Secs. VII and VIII show is jointly excluded by its own tensor-to-scalar ratio and this same non-Gaussianity: no constant sound speed relieves both at once. Conditional on that background, the amplitude is not contradicted by any channel currently able to speak, but three of the four channels tested are honest nulls rather than supporting consistencies. Pulsar timing: propagating this model's own curvature spectrum through the validated induced-GW kernel gives $\gamma_{\text{pred}} = 5.07$ (5.00, dust bracket), a Gaussian-equivalent 5.1σ (4.9σ) z -distance from NANOGrav's measured $\gamma_{\text{HD}} = 3.2 \pm 0.6$, at an amplitude $10^{15.2}$ below the NANOGrav band (Sec. IVD); the $\gamma = 3$ slope quoted for a matter bounce elsewhere in the literature [8] requires a small-scale spectral enhancement this model does not have, and is retained only as a flagged contrast case. This channel is also insensitive to the value of f_{NL} and therefore does not test it either way. Primordial black holes yield a reproducible amplitude ratio (1.84 ± 0.03 within this model's own γ_{cr} coverage) between the two candidate values, but the lab's own predicted spectrum falls 6.7–7.0 orders of magnitude short of the required amplitude at every mass scale considered (Sec. VC): the channel is a clean null for this model, not an open promise. The same lab spectrum falls three orders of magnitude short of the amplitude an early-supermassive-black-hole-seed channel would require (Sec. VC): a third honest null. Transmission through the bounce is now a definite prediction rather than an open bound: at $k\eta_B \lesssim 10^{-2}$ on the Quintin-type background, the scheme ambiguity is resolved by the Bardeen-potential argument of Sec. III A in favor of S2, giving the single value $f_{\text{NL}}^{\text{after}} = -1.25$; on the LQC and poly backgrounds the same Bardeen continuation now gives a linear-transfer amplitude of -1.094 and -1.140 respectively (the S1 values, $\in [-0.65, -0.55]$, are superseded but retained for comparison), with the cubic term not yet computed there, so the full $f_{\text{NL}}^{\text{after}}$ is only a bracket, $[-1.20, -0.09]$ and $[-1.27, -0.13]$ (Sec. III A). As Sec. III states, this $k\eta_B$ window is an upper bound on k and is satisfied *most easily*, not *hardest*, at the CMB/LSS pivot scale used by Channel III below, so these transmitted values are precisely what that channel should be compared against. Galaxy surveys are the only channel that discriminates between the two candidate amplitudes at all, but at the transmitted amplitude that discrimination is weak: DESI DR1 is consistent with $f_{\text{NL}}^{\text{after}}$ at well under 1σ under either prior, and SPHEREx reaches only 0.7 – 0.9σ ($-35/16$) or 1.2 – 1.7σ ($-35/8$) bare significance (Table VI), and combining power spectrum

and bispectrum widens the candidate separation to only $0.5\text{--}1.1\sigma$ at current or forecast LSS sensitivity, not the 3.13σ forecast detection significance quoted for the pre-bounce amplitude a naive reading of Channel III alone would suggest: the two figures are *not directly comparable*, since 3.13σ assumes $T_{f_{\text{NL}}} = 1$ (Table VI’s clearly-labelled secondary row), not the transmitted amplitude this paper otherwise supports. No channel observationally contradicts $f_{\text{NL}}^{\text{local}} = -35/16$; the honest summary is three nulls (pulsar timing, PBH abundance, SMBH-seed abundance) and one channel (large-scale structure) that is reachable in principle but does not yet separate the two candidates.

B. Where the consistency is weak

The pulsar-timing channel is now a definite null rather than an open consistency: as Sec. IVD shows, this model’s own spectrum predicts $\gamma_{\text{pred}} \approx 5$, not the $\gamma = 3$ associated elsewhere in the literature with a small-scale-enhanced matter-bounce source [8] this model does not have – and it is identical at both candidate f_{NL} values regardless. The black hole channel now uses the literature’s own formation criterion and the lab’s own spectrum rather than an external, unreconstructible one, and returns a definite verdict (null) rather than an open amplitude scan; the amplitude ratio remains the channel’s only shape-robust, spectrum-independent output, and does not by itself separate $-35/16$ from $-35/8$ observationally – only compares how hard each is to produce – and $f_{\text{PBH}}(f_{\text{NL}})$ itself was shown to be strongly non-monotonic along the path to both candidates on the earlier lognormal grid (Sec. VB). The transmission result is validated only for $k\eta_B \lesssim 10^{-2}$; on the Quintin-type background the scheme ambiguity in the linear transfer is resolved (Sec. III A): a scheme-independent Bardeen-potential argument selects the effective-fluid scheme S2, whose raw (pre-integration-by-parts) cubic action, evaluated on its exact mode functions, is finite – its apparent divergence in the integrated-by-parts form is a total-derivative artefact, not a physical pathology – and gives the single value $f_{\text{NL}}^{\text{after}} = -1.25$, a factor ≈ 2.5 from the earlier S1 value -0.50 on the same background (Sec. III A). Which scheme’s linear variable is physical at $H = 0$ is therefore settled on this background; the same continuation now extends to the LQC and poly backgrounds, where the crossing is smooth rather than a jump, but there the continuation carries an $O(1)$ self-adjoint-extension freedom the Quintin-type jump avoids – quantified and, on LQC only, independently resolved (Sec. III A) – and the cubic term remains uncomputed, so only the linear-transfer values and a bracket are reported (Table III). At the transmitted amplitude the survey channel’s discriminating power is itself the weak link: SPHEREx separates the two candidates by $|\Delta f_{\text{NL}}^{\text{after}}| \approx 0.35\text{--}0.55$ against $\sigma_{f_{\text{NL}}} = 0.5\text{--}0.7$, i.e. $0.5\text{--}1.1\sigma$ (Sec. VI), so no currently forecast LSS survey can distinguish $-35/16$ from $-35/8$

once scheme-S1 transmission is applied; a not-yet-derived shape-overlap projection $r_{\text{sh}} < 1$ would only narrow this separation further, and since r has not been computed at this paper’s fiducial no quantitative statement about its effect is made here.

Four further limitations are stated here rather than answered. (a) The S1 prescription ($z = a$, $\epsilon_{\text{eff}} = 1/2$) is carried unchanged to $c_s \neq 1$, where the $1/c_s^4$ growth of Eq. (17) originates in boundary terms and the $\zeta\dot{\zeta}^2$ vertex; no independent (S2 or exact-mode) evaluation supports that scaling below $c_s \approx 0.6$, which is one reason the window’s small- c_s end is reported as a validity boundary rather than a value (Sec. VIII). (b) Assumption (A4) freezes the contraction bispectrum at η_h and evolves it linearly, although the growing mode still has $\dot{\zeta} \neq 0$ there; the resulting evaluation-time ambiguity is not quantified against the width of the S1 three-background band $[-0.65, -0.50]$. (c) The comoving-slice δN value $f_{\text{NL}}^c = -5$ is well-defined as a variable, but the gradient expansion underlying any separate-universe construction is itself invalidated by the $N_i = O(1/k_L)$ shift contribution identified in Appendix A, so its status as a physical prediction – rather than as one member of an exactly-related family of variables – is not settled here. (d) The sign and coefficient of $\Delta f_{\text{NL}}^{\text{bounce}}$ are computed for a window symmetric about the bounce; robustness to an asymmetric window is untested.

C. Next steps

In priority order: (i) computing the cubic-order raw-ADM term on the LQC and poly backgrounds under the now-extended Bardeen continuation (Sec. III A), whose $Q = 0$ crossing carries the same logarithmic structure at cubic order, so the reported bracket (Table III) can be replaced by a single number; relatedly, resolving the self-adjoint-extension ambiguity of the linear continuation there – on poly by finding a prescription-free anchor analogous to LQC’s or an independent argument fixing the physical choice, and on LQC by testing whether the effective theory’s kinetic weight indeed leaves the anchor prescription-free; (ii) an evaluation-time (freeze-out) prescription for the bispectrum at $k\eta_B \gtrsim 0.1$, where this model’s absence of a post-bounce inflationary phase leaves ζ unfrozen and the bounce-window cubic integral dependent on when it is evaluated by a factor of several, so that the $k\eta_B \approx 0.6\text{--}0.8$ transfer feature and the $k\eta_B \sim 1\text{--}3$ exact-mode bispectrum of Sec. III can be quoted as single numbers rather than ranges; (iii) re-derivation of the shape overlap r at the $-35/16$ fiducial; (iv) extending the S2 raw-ADM computation to the $-35/8$ amplitude on all three backgrounds, not computed here (Sec. III A, Table VI); and (v) a direct-collapse (rather than horizon-re-entry) primordial-black-hole formation channel during the dust contraction phase itself, identified but not evaluated in Sec. VC. The propagation of the post-bounce amplitude to the CMB/LSS pivot and of the scalar spectrum

through the induced-GW kernel to the nHz amplitude are carried out in Secs. VI and IV D; the bounce-scale enhancement mechanisms at $k\eta_B \sim 1$ that would be the remaining route to reopen Channel I or the black-hole channel have been checked directly (velocity dip, transfer excursion, and LQC cubic-operator enhancement, Sec. III) and found not to reopen either. Two further items follow directly from Secs. VII–VIII: (vi) converting the curvaton route of Sec. VIII into a concrete, checkable prediction requires an entropy-sector kinetic-amplification mechanism realized on the A2 bounce backgrounds used throughout this paper, and a demonstration that σ_* stays constant across them – neither is modelled here, and Cai, Brandenberger & Zhang’s [25] own ζ_{NL} integral through deflation and bounce remains un-re-derived by this lab; and (vii) the consistency relation $n_T = n_s - 1 = -0.035$ (Sec. VII) is a surviving, in-principle discriminator of this class of background from single-field inflation independent of the (r, f_{NL}) no-go, worth a dedicated forecast. The general- P_{XXX} coefficient λ does not weaken the result: Sec. VIII shows $r = 24c_s$ and $\Delta f_{NL}^{\text{bounce}}(c_s)$ are exactly λ -independent, so the no-go holds for the full $P(X)$ class (reproducibility statement below).

Appendix A: The δN_c – ζ_{Mald} derivation

This appendix transcribes, without new science, the derivation archived in the note cited in the reproducibility statement below (§§1–4) that underlies Eq. (5) and the [L]/[K]/[X]/[S] decomposition of Sec. II D.

1. The linear-order identity, Eq. (5)

On a uniform- ϕ slice the lapse is $N = 1 + \dot{\zeta}/H$, so the Friedmann equation at fixed ϕ gives $\delta H/H = -(\epsilon/3)\dot{\zeta}_L/H$: the local expansion rate on comoving slices is perturbed at $O(k^0)$, carried by the trace of the shift gradient, $\delta K = -\partial^2\psi/a^2 \rightarrow -\epsilon\dot{\zeta}_L$. The separate-universe δN is defined on the zero-shift threading, $\delta N_c = \int (K/3) d\tau$, so writing $\delta N_c = \zeta + b$ with $\dot{b} = \delta K/3$ gives, on the growing mode, $b = -(\epsilon/3)\zeta_L$, i.e. Eq. (5), $\delta N_c = (1 - \epsilon/3)\zeta_{\text{Mald}}$ at linear order (sign convention: ζ_{Mald} and δN_c share the sign of the comoving-gauge curvature perturbation used throughout this paper).

2. The [L]/[K]/[X]/[S] decomposition

Separating the long mode’s coupling to the short-mode pair by how it enters:

class	long mode enters as	squeezed value at $\epsilon = 3/2$
[L]	$\zeta_L, \dot{\zeta}_L$ (lapse/curvature; local response)	$-25/8$, pure monopole
[K]	$\partial_i\partial_j\tilde{\chi}_L$ (δK and shear)	0 (cancels between varied legs of T_4)*
[X]	$\partial_i\tilde{\chi}_L$ (the <i>shift</i> , $1/k_L$)	$\propto \frac{15}{16} + \frac{15}{16}\mu^2$ (monopole $5/4$)
[S]	short-mode sourcing of the long mode	0 ($O(k_L^2/a^2H^2)$ -suppressed)
total		$-\frac{35}{16} + \frac{15}{16}\mu^2$ (= in-in exactly)
comoving δN	different variable, no [X]	-5 , isotropic

*Here T_3 and T_4 denote the cubic- and quartic-leg vertex structures of the comoving-gauge action, i.e. the terms obtained by expanding the ADM action to third and fourth order in ζ before the constraint solution for the lapse and shift is substituted; [K] receives equal and opposite contributions from the two varied legs of T_4 .

The $1/k_L$ poles in [X]’s two contributions – the T_3 bulk coupling and the boundary term f_b below – cancel only in their sum, leaving a finite $O(k_L^0)$ remainder; no separate-universe construction (isotropic or Bianchi-I) contains this term, because it originates in the comoving-gauge *shift* $N_i = \partial_i\psi \supset a^2\epsilon\partial^{-2}\zeta$, which is $O(1/k_L)$ in this non-attractor phase and so violates the gradient-expansion assumption $N_i = O(\nabla)$ every separate-universe construction relies on. The final-time re-threading boundary term,

$$f_b = \frac{\epsilon}{2\mathcal{H}} \left[\partial\zeta \partial\tilde{\chi} - \partial^{-2}\partial_i\partial_j(\partial_i\zeta \partial_j\tilde{\chi}) \right], \quad (\text{A1})$$

subtracts $(1 - \epsilon/3) \times \frac{5\epsilon}{4}\mu^2$ from the T_3 bulk’s μ^2 coupling and adds the $+15/16$ monopole, reproducing the μ^2 coefficient exactly and supplying the [X] monopole. In-in monopole minus δN_c : $-\frac{15}{8} - (-5) = \frac{25}{8} = [X]_{\text{mono}} (5/4) + ([L] - \delta N_c) (15/8)$; in general ϵ , $5\epsilon(9 - \epsilon)/18 = 5\epsilon(9 - 2\epsilon)/36 + 5\epsilon/4$. The zero-shift, fluid-worldline threading identity underlying δN_c is $\delta N_c(x_f) = \zeta(t_f, x_f) - \frac{1}{3} \int \partial_i N^i dt$ along the worldline; expanding it to second order at constant ϵ gives $f_{\delta N_c} = f^{\text{in-in}}/\lambda + f_{\text{map}}$ with $\lambda = 1 - \epsilon/3$. The worldline can be labelled by its position on the *initial* slice (the label the separate universe uses, since it lays out its patches there) or on the *final* slice; the two maps are

$$f_{\text{map}}^{\text{init}} = \frac{5\epsilon}{4(3 - \epsilon)} [(\epsilon - 2) - \epsilon\mu^2], \quad f_{\text{map}}^{\text{fin}} = -\frac{5\epsilon}{4}(1 - \mu^2), \quad (\text{A2})$$

each with monopole $-5\epsilon/6$ at $n_s = 1$, the value used throughout this paper’s $\epsilon = 3/2$ dust construction (a general-tilt gate finds the final-label map monopole is $-5\epsilon/6$ for any n_s , while the initial-label one is $\frac{5\epsilon(2\epsilon - 7 + n_s)}{12(3 - \epsilon)}$, which reduces to $-5\epsilon/6$ only at $n_s = 1$; reproducibility statement below). The *initial-label* composition, $f_{\delta N}^{\text{init}} = f^{\text{in-in}}/\lambda + f_{\text{map}}^{\text{init}}$, is an isotropic monopole

of exactly -5 for every constant ϵ , not a numerical coincidence; this is the label the separate universe computes. The *final-label* composition, $f_{\delta N}^{\text{fin}} = f^{\text{in-in}}/\lambda + f_{\text{map}}^{\text{fin}} = -\frac{25}{4} + \frac{15}{4}\mu^2$ at $\epsilon = 3/2$, differs by a translation term with no monopole; the monopole gap to the in-in value is label-independent. The relation between δN_c and ζ_{Mald} is an exact, invertible change of variable ($f_{\delta N_c} = f^{\text{in-in}}/\lambda + f_{\text{map}}$, $\lambda \neq 0$); the $O(1)$ effect above is an error only in the standard δN formula's identification $\delta N_c = \zeta$, and it arises iff the neglected super-Hubble integral $I = \zeta_{L,f}^{-1} \int_{t_i}^{t_f} (\epsilon/c_s^2) \dot{\zeta}_L dt$ is $O(1)$ – not from any physical translation mechanism.

3. General- ϵ formulas

The comoving separate-universe results, for general ϵ , are

$$f_{\text{NL}}^\rho = \frac{5(\epsilon - 7)}{8}, \quad f_{\text{NL}}^c = -5 \quad (\text{all } \epsilon), \quad (\text{A3})$$

and the in-in squeezed bispectrum, general ϵ at $n_s = 1$, is

$$f^{\text{sq}}(\mu, \epsilon) = \frac{5}{12}(\epsilon^2 \mu^2 - \epsilon^2 + 6\epsilon - 12), \quad (\text{A4})$$

with monopole $-5(\epsilon-3)(\epsilon-6)/18$, isosceles ($\mu = 0$) value $-5(\epsilon^2 - 6\epsilon + 12)/12$, and μ^2 coefficient $5\epsilon^2/12$; at $\epsilon = 3/2$ these reduce to the monopole $-15/8$, the isosceles value $-35/16$ of Eq. (4)'s $\mu = 0$ limit, and the $+15/16 \mu^2$ coefficient (Legendre $\ell = 2$ coefficient $5/8$) quoted throughout this paper. A constant- ϵ background is scale-invariant only at $\epsilon = 3/2$ (the dust case used throughout this paper) and, on the dominant super-horizon mode, as $\epsilon \rightarrow 0$; away from those points $n_s \neq 1$ and f^{sq} carries an extra $-\frac{5}{12}\epsilon(n_s - 1)\mu^2$, but a general-tilt gate finds this correction cancels identically against the equal and opposite shift in $f_{\delta N}^{\text{init}}$'s monopole after the $1/\lambda$ composition above, so $f_{\delta N}^{\text{init}} = -5$ for every constant ϵ and every n_s (reproducibility statement below); no number in this section or Sec. IID, all evaluated at the dust value $\epsilon = 3/2$, is affected. These general- ϵ expressions hold on the non-attractor growing branch ($\dot{\zeta} = -3\zeta/\eta$) that this paper's entire construction uses, and do *not* reduce to the slow-roll attractor (Maldacena) result $(5/12)(1 - n_s) \rightarrow 0$ as $\epsilon \rightarrow 0$; they are not claimed to.

4. Bianchi-I: why an anisotropic separate universe does not close the gap

An anisotropic (Bianchi-I) separate-universe treatment, carrying the long mode's projected shear onto the short-mode power spectrum exactly, gives an identically zero monopole contribution: the shear is $\mathcal{O}(k_L^0)$ and traceless, and no traceless linear-in- ζ_L response can supply a monopole. It therefore cannot by itself

close the $-25/8$ gap between $\delta N^c = -5$ and the in-in monopole $-15/8$; the gap is closed only by [X], the shift term no separate-universe construction contains (Sec. A2 above). A direct second-order continuation of the uniform-density (ρ -slice) δN variable itself – as opposed to a Bianchi-I correction to the isotropic construction – gives $f_{\text{NL}}^\rho = 5(2\epsilon - 15)/24 = -5/2$ at $\epsilon = 3/2$ (isotropic, initial label); this does not reproduce the separate-universe value $-55/16$ quoted in Sec. IID (gap $15/16$). The intrinsic term, $f_{\text{NL}}^{\text{intr}} = \frac{5\sqrt{3}\epsilon(3-\epsilon)}{18} \frac{r_i}{W-1}$, vanishes in the growing-mode limit that defines $-55/16$ for any t_f -independent r_i , so it does not close this gap. An independent exact separate-universe solution reproduces both $-55/16$ (uniform-density slice) and -5 (comoving slice) for all ϵ , so the uniform-density δN value is itself well-defined, not an evolution-step artifact; the gap $5(6 - \epsilon)/24$ is closed, per an independent adjudication (reproducibility statement below), by correcting the composition weight on the second-order curvature perturbation from $\lambda' = 2\lambda$ to 3λ : the threading map's long $\times$ short lapse monopole coefficient $A_2 = \epsilon(3 - \epsilon)^2/3$ (this map) is the correct constraint-solve value, not the separate universe's $2(3 - \epsilon)^2$. Both the comoving-slice δN value (-5) and the uniform-density δN value ($-55/16$) are now fully reconciled by the map derived here; the flagship in-in monopole $-35/16$ is unaffected.

REPRODUCIBILITY STATEMENT

Every result reported here traces to an archived script, chain, or reproducibility manifest; nothing is quoted from memory. All artifacts named below live on the `main` branch of the public repository, at

https://github.com/Hubify-Projects/bigbounce/tree/main/research/theory_audit

https://github.com/Hubify-Projects/bigbounce/tree/main/research/cubic_bounce_transmission

https://github.com/Hubify-Projects/bigbounce/tree/main/research/track_a3_multichannel

covering the exact-amplitude derivation (Sec. II), the bounce-transmission bound (Sec. III) and the channel computations respectively, and including `fnl_monopole_adjudication_2026_09_03.md` in the first (Appendix A). The general-tilt gate on Appendix A's monopole and $f^{\text{sq}}(\mu, \epsilon)$ expressions is `psu_gate_S12_translation_trace_2026_09_19` in the first location, manifest `psu_gate-s12-translation-trace`. A branch pointer is used deliberately: several of the artifacts below postdate any single earlier commit, so no one historical hash resolves to all of them. The immutable pointer for this manuscript is the frozen-release DOI that will be deposited at publication (see the end of this statement); the branch trees above are the pointers guaranteed to resolve in the meantime. The mechanisms of Sec. III and the S2 resolution of Sec. IIIA are under `research/cubic_bounce_transmission/`:

lane9a_velocity_dip/ (Quintin Eq. 79 evaluation and the $k\eta_B \in [0.1, 10]$ transfer scan), lane9b_s2_regulation/ (the total-derivative diagnosis of the S2 divergence), lane9b2_s2_rawadm/ (the raw-ADM S2 computation, results.json), lane9c_abs_operator/ (the Agullo–Bolliet–Sreenath operator map and scale-window conversion), and lane9c2_lqc_modes/ (the exact-mode LQC-dust in-in evaluation, results.json), each with its own manifest under reproducibility/manifests/experiments/. The Bardeen-potential scheme selection of Sec. III A (the Φ/Ξ first-order system, the $|\lambda_c| = 0.9699$ propagation, the smoothed-NEC-crossing control, and the independent blind adjudication) is at research/cubic_bounce_transmission/row9_scheme_independence_2026-09-19/ (row9_symbolic.py, row9_numeric.py, results.json, symbolic_results.json), manifest a3-row9-scheme-independence. The tensor $H = 0$ continuation of Sec. VII is at research/cubic_bounce_transmission/row18a_s2_tensor/ (results.json); the c_s -dependent bounce cubic term of Sec. VIII is at research/cubic_bounce_transmission/row18b_cs_bounce_cubic/ (row18b_cs_bounce_cubic.py, results.json). The general- λ extension of Sec. VIII (the λ -independence of r and $\Delta f_{\text{NL}}^{\text{bounce}}$, and the λ/Σ scan) is at research/track_a3_multichannel/row19_lambda/ (row19_lambda.py, results.json, row19_lambda.log), manifest reproducibility/manifests/experiments/a3-row19-lambda.json. The perturbativity boundary of Sec. VIII (the c_s floor at which $1.2|f_{\text{NL}}^{\text{after}}|_{\zeta_{\text{rms}}} = 1$, and the 68%/95% Planck edges of Table VII) is at research/track_a3_multichannel/r9_perturbativity/ (r9_perturbativity.py, results.json); it reproduces the 68% boundary $c_s \geq 0.624/0.632/0.627$ quoted in Sec. VIII as an independent check. The criterion-explicit Λ -line window minima of Sec. VIII (the Planck 95% and 68% edges for the DBI, $P \propto X^n$ and $\Lambda = 0$ lines on all three backgrounds) are at research/track_a3_multichannel/r10_window_criterion/ (r10_window_criterion.py, results.json), and the per-row evaluation of Table V at its own labeled $(\Delta, r_p k_p, C_{\text{th}})$ with per-row Gaussian calibration is at research/track_a3_multichannel/r10_tableV_recompute/ (r10_tableV_recompute.py, results.json), together with the measured ratio trends quoted above – the ordinary-least-squares slope in γ_{cr} , the C_{th} step, and the family composition of the 144-point subset – in r10_ratio_slope.py, ratio_slope.json. The step-convergence figures quoted in Table III’s caption are in research/cubic_bounce_transmission/lane_b_numerical/ (results.json). The bounce-to-local shape overlap ($r_{\text{sh}} = 1$, an assumed value pending an explicit shape-overlap projection; Sec. VI) is taken from this lab’s companion f_{NL} -forecast program (outputs/survey_reach_fnl.json); every tabulated

survey significance in Table VI is accordingly an upper bound on the detectable significance, not a forecast of it. The three channel computations of Secs. IV–VI are archived at research/track_a3_multichannel/: pta_gamma_reproduce.py, pbh_abundance_fnl.py (first-pass Press–Schechter, retained for comparison only), pbh_compaction_fnl.py (compaction-function criterion, this paper’s principal PBH result), inlab_delta2_zeta_2026-09-03.py (Sec. VC, the lab’s own predicted spectrum and its FIRAS check), survey_reach_fnl.py, pta_injection_30bin_2026-09-02.py, and pta_injection_30bin_realkde_2026-09-02.py (the Sec. IVC injection-recovery validations), and PBH_COMPACTIION_NOTE_2026-09-02.md (the Sec. VB compaction-note validation), and sigw_nhz_from_lab_spectrum_2026-09-04.py (Sec. IVD, the induced-GW SIGW computation), with results in outputs/*.json and manifests at reproducibility/manifests/experiments/a3-*.json (PTA/PBH/reach, including a3-pbh-compaction-fnl.json), reproducibility/manifests/experiments/p2-fnl-*.json (exact-amplitude derivation and validation), and reproducibility/manifests/experiments/p2-a2-*.json (bounce-transmission computation). The pulsar-timing chain is pipelines/p3_pta_mcmc/free_spectrum_real_2026-05-01/chain_real_freespec.npy (the directory date is this lab’s run date of the reduction, not the underlying NANOGrav release date) (SHA-256, in two halves to allow line-breaking: 50abc38a04e1bf886adc833b5eb653be4e986877fe2476f87a78927f4f3610fc), derived from the NANOGrav 15-yr KDE free spectra, Zenodo 10.5281/zenodo.8060824. This work’s own code and artifacts are referenced by the branch trees above; a frozen-release DOI covering them will be deposited at publication. The exact-amplitude theory lineage this paper builds on is archived at Zenodo 10.5281/zenodo.21461881. The DESI DR1 QSO reproduction of Sec. VI is archived at research/desi_png_reproduction/, manifest LEDGER4_RESULT_v3_2026-09-04.md. The tensor amplitude of Sec. VII, including the $\gamma_{\text{pred}}^{(1)} = 5.035$ nHz propagation of Sec. IVD (outputs/r5_15_tensor_omega_nhz.json), is archived at research/track_a3_multichannel/row10_r_ns/ (row10_r_ns.py, results.json); the joint (r, f_{NL}) no-go of Sec. VIII is at research/track_a3_multichannel/row14_cs_window/ (row14_cs_window.py, results.json); the curvaton-route calculation is at research/theory_audit/curvaton_matter_bounce_adjudication_2026-09-04.py; the PBH sign-correction and extended γ_{cr} scan of Sec. VB, including results/row11_gammacr_extension.json, is at research/track_a3_multichannel/

row11_pbh_residuals/; and the Appendix A threading-map identity is at `research/theory_audit/threading_map_second_order_2026_09_04.py`, each with a manifest under `reproducibility/manifests/experiments/`. Every computation reported in this paper ran on a single laptop CPU at zero compute cost. Wall-clock times: pulsar timing 0.02s, survey reach 0.002s, induced-GW spectrum 2.7s, in-lab curvature spectrum 49s, PBH compaction grid 215s, and the extended γ_{cr} scan 1283s. Compute venue, wall-clock time, and external data sources are recorded per-experiment in the manifests cited above.

AI USAGE DISCLOSURE

Symbolic derivations reported in Secs. II–III were performed by deterministic sympy scripts written and exe-

cuted under the author’s direction; every intermediate result is printed by the archived scripts and independently checked against the cited literature limits before use. The channel computations of Secs. IV–VI were performed by archived Python scripts against real data and cited published values, not by an LLM. Claude (Anthropic) assisted with literature cross-checks, prose drafting, and L^AT_EX formatting under the author’s direction; all physics content, equations, and numerical results were verified by the author against the archived script output.

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